

ForPrint Human Intent Ledger + Expanded Portfolio Projection v0.2

Generated 2026-08-31

Authority note: this PDF is a human review projection; structured Blueprint governance, roadmaps, contracts, ledgers and evidence remain authority.

Human Intent coverage: 577 / 577 indexed IDs.

Open GAP list: [coordination/internal_work/blueprint/evening_reviews/2026-08-31/open_gaps_and_deferred_decisions_v0_1.md](#)

Exact Calculator external visual-configurator resources/URLs remain unrecovered. Do not invent replacements.

Module — forprint_system_blueprint

Intent count: 215

AGREED · HI-FP-SYSTEM-BLUEPRINT-001

Після кожного великого вечірнього діалогу має з'являтися не лише технічний пакет інтеграції, а й окремий людський список домовленостей по кожному модулю.

Context: Це нове правило сформульоване сьогодні, щоб не втратити сенс розмови під час технічної нормалізації.

AGREED · HI-FP-SYSTEM-BLUEPRINT-002

Людські тези мають накопичуватися append-only: нові записи додаються, старі не зникають; якщо думка змінилася, стара теза позначається superseded, але лишається видимою.

Context: Портфоліо повинно показувати історію наміру, а не лише поточну машинну формулу.

AGREED · HI-FP-SYSTEM-BLUEPRINT-003

Для кожного roadmap step потрібне посилання на human-intent записи, які пояснюють, навіщо цей крок існує і що саме люди мали на увазі.

Context: Це дає можливість повернутися до контексту при двозначності реалізації.

AGREED · HI-FP-SYSTEM-BLUEPRINT-004

Потрібно підтримувати три представлення портфоліо: вузьке машинне, збалансоване технічне та велике людське українською мовою.

Context: Велике портфоліо не повинно стискати десятки домовленостей до однієї загальної фрази.

AGREED · HI-FP-SYSTEM-BLUEPRINT-005

Один front door для нового помічника має вести до current authority, roadmap, standards, knowledge indexes, active work і потрібного context bundle.

Context: Без задачі даємо широкий bootstrap; з конкретною задачею — вузький task bundle.

AGREED · HI-FP-SYSTEM-BLUEPRINT-006

make check залишається детермінованим і read-only: він може сказати INDEX_DRIFT, але не повинен тихо переписувати індекси чи завершувати роботу за людину.

Context: Автоматичні виправлення не повинні ховатися всередині перевірки.

AGREED · HI-FP-SYSTEM-BLUEPRINT-007

Кожна робота стартує на pinned governance/process revision і здається проти цієї ревізії, якщо вона не була явно BLOCKED/REVOKED.

Context: Нова ревізія не повинна ретроспективно ламати вже розпочату роботу без явної причини.

AGREED · HI-FP-SYSTEM-BLUEPRINT-008

Completion attestation має містити pinned snapshot, standards versions, context bundle, checks, evidence paths, deviations і стан roadmap/contracts.

Context: Мета — довести відповідність, а не просто написати 'помічник прочитав стандарти'.

AGREED · HI-FP-SYSTEM-BLUEPRINT-009

Inspector повинен незалежно перевіряти completion reports і drift, а Blueprint — показувати портфельну картину та баланс між модулями.

Context: Inspector не стає semantic owner модулів.

RECOVERED · HI-FP-SYSTEM-BLUEPRINT-010

Blueprint має бачити critical path, blockers, WIP, progress weight, cost/quality/executor evidence і балансувати рух модулів.

Context: Це вже було в попередньому розширеному портфоліо.

AGREED · HI-FP-SYSTEM-BLUEPRINT-011

PDF-портфоліо є projection для людського review, а не authority. Authority — структуровані записи, roadmap і governance surfaces.

Context: Щоб дизайн документа не ставав єдиним місцем, де живе домовленість.

AGREED · HI-FP-SYSTEM-BLUEPRINT-012

Canonical UI design system має бути reusable across Website/Mobile/internal surfaces; page-local visual systems не повинні множитися.

Context: Library owner, consumers adopt.

AGREED · HI-FP-SYSTEM-BLUEPRINT-013

Human portfolio має показувати dependency of each step, а не лише функціональність: хто має закінчити що, щоб цей модуль міг рухатися далі.

Context: Це основа балансування модулів.

AGREED · HI-FP-SYSTEM-BLUEPRINT-014

Expanded portfolio rendering/content rules must be documented so the same recognizable PDF structure can be regenerated by another assistant or generator.

Context: Portfolio layout must not depend on conversational memory.

Roadmap: BP-PORTFOLIO-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-015

Portfolio roadmap should show the rough full route to target state even when distant steps are not yet fully discussed.

Context: Future steps may be synthetic if they are visibly marked and do not pretend to be approved implementation.

Roadmap: BP-PORTFOLIO-02

AGREED · HI-FP-SYSTEM-BLUEPRINT-016

Synthetic portfolio content is useful and should remain visible with a distinct status/color plus confidence/evidence basis.

Context: Synthetic does not mean bad; it means useful forward projection that still needs review.

Roadmap: BP-PORTFOLIO-03

AGREED · HI-FP-SYSTEM-BLUEPRINT-017

Historical portfolio PDFs must remain in the project under versioned/date filenames and must not be overwritten by a newer projection.

Context: The PDF is history/projection, not release authority.

Roadmap: BP-PORTFOLIO-04

AGREED · HI-FP-SYSTEM-BLUEPRINT-018

Cross-module semantic invariants should become a machine-readable governance surface consumed by Inspector.

Context: Examples include phone-not-identity, QR-not-permission and filename-not-source-of-truth.

Roadmap: BP-SEMANTIC-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-019

Important operator wording that explains why a roadmap step exists must be preserved as curated verbatim quotes linked to Human Intent and roadmap context.

Context: Preserve selected exact wording, not a full chat archive.

Roadmap: BP-PORTFOLIO-05

AGREED · HI-FP-SYSTEM-BLUEPRINT-020

Every module roadmap must show a visible route to a mature Final Target State split into human-confirmed and synthetic/proposed capabilities.

Context: If a useful mature purpose cannot be formulated even synthetically, reassess whether the module should exist.

Roadmap: BP-PORTFOLIO-06

AGREED · HI-FP-SYSTEM-BLUEPRINT-021

Ordinary module implementation remains held until full-horizon roadmaps, module self-inventory and a dependency/readiness/priority dashboard are sufficiently complete.

Context: Planning, inventory and Blueprint tooling remain allowed.

Roadmap: BP-PORTFOLIO-07

AGREED · HI-FP-SYSTEM-BLUEPRINT-022

Meaningful new capabilities must be checked against project capability/index surfaces before implementation to reduce duplicate code and semantic drift.

Context: Self-inventory is an architecture control, not only documentation.
Roadmap: BP-SEMANTIC-02

AGREED · HI-FP-SYSTEM-BLUEPRINT-023

ForPrint uses one central PostgreSQL operational platform with logical domain schemas, stable shared IDs, controlled writes and reporting projections.

Context: Physical centralization does not mean shared semantic ownership.
Roadmap: BP-DATA-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-024

ForPrint Control Center uses left-side module navigation and top tabs within the selected module workspace.

Context: Cloud Backup Manager is a reference shell; shared Library UI rules remain authoritative.
Roadmap: BP-UI-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-025

Deterministic-to-AI fallbacks should be logged and periodically reviewed for conversion into safe deterministic automation.

Context: Do not repeatedly pay for the same solvable automation gap.
Roadmap: BP-AUTOMATION-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-026

Project cleanliness is a permanent Blueprint architecture objective: recurring current document and machine-surface classes should converge on registered canonical schemas, generator/source relationships and validators instead of accumulating format variants.

Context: Normalization must preserve semantics and migrate legacy debt in controlled class-by-class batches; new unregistered structural debt should not be allowed to grow.
Roadmap: BP-CLEAN-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-027

Evening/morning architecture handoff information must first be absorbed into the Blueprint portfolio, Human Intent and module roadmaps; module assistants are not contacted or started while this internal analysis is still being completed.

Context: Deepen the personal Blueprint project first; distribution is a later phase.
Roadmap: BP-PORTFOLIO-08

AGREED · HI-FP-SYSTEM-BLUEPRINT-028

Assistant task distribution becomes eligible only after full portfolio analysis, roadmap/dependency review and explicit operator approval; planning packages do not themselves authorize execution.

Context: No prompt activation, module implementation, ACCEPT/RETURN/HOLD, H10 widening, commit or push is implied.
Roadmap: BP-PORTFOLIO-09

AGREED · HI-FP-SYSTEM-BLUEPRINT-029

Project cleanliness must be a permanent project-wide policy: Blueprint keeps the single canonical cleanliness rules, every module is responsible for cleanliness of its own repository, and modules must not create competing global standards or uncontrolled document/surface variants.

Context: Operator explicitly requires the cleanup procedure learned in Blueprint to become a general rule for all modules.
Roadmap: BP-PORTFOLIO-08

AGREED · HI-FP-SYSTEM-BLUEPRINT-030

Automated cleanliness capability must be discoverable and status-explicit: existing Blueprint checks are recorded as implemented; missing module-local and cross-repository controls are recorded in portfolio roadmaps before future execution.

Context: Do not describe planned automation as already implemented; keep implemented evidence and future work separate.
Roadmap: BP-PORTFOLIO-09

AGREED · HI-FP-SYSTEM-BLUEPRINT-031

Project Inspector is the cross-repository cleanliness/conformance auditor, while each module retains local semantic ownership and local repository cleanliness responsibility.

Context: Inspector detects/report drift and duplicates but must not silently rewrite domain truth.
Roadmap: BP-PORTFOLIO-09

AGREED · HI-FP-SYSTEM-BLUEPRINT-032

Before future module assistants are started, Blueprint must define and review an AI Execution Safety and Runtime Governance gate covering ingress validation, tool-call validation, egress/data-release control, bounded retries/loops, cost budgets, execution logging and manual escalation.

Context: A raw Telegram message or arbitrary AI request must not automatically become trusted executable work.
Roadmap: BP-AI-SAFETY-01

AGREED · HI-FP-SYSTEM-BLUEPRINT-033

Retry, clarification, inter-module, database, tool, AI/token/cost and time limits must be configurable by operation/risk class rather than one universal fixed retry count.

Context: Repeated identical invalid calls must terminate or escalate instead of entering an unbounded loop.
Roadmap: BP-AI-SAFETY-02

AGREED · HI-FP-SYSTEM-BLUEPRINT-034

Before the next detailed portfolio is approved, older and under-documented modules must be reconciled against their repository inventories so historical capabilities, stale instructions and misowned functionality are not silently carried into the mature architecture.

Context: Calculator and Telegram are especially important because they existed before most current domain modules.
Roadmap: BP-PORTFOLIO-10

AGREED · HI-FP-U92-20260903-009

The mature ecosystem should reduce routine human steering to a small number of control surfaces while modules operate under bounded, reversible managed autonomy.

AGREED · HI-FP-U92-20260903-013

AI-generated commands are not authorization: side effects must pass a deterministic Execution Policy Gate, IAM/resource policy and capability-shaped tool boundary.

AGREED · HI-FP-U92-20260903-014

Development, Verification, Release and Runtime are separate execution profiles; runtime deployed code must not self-modify live source or install arbitrary dependencies.

AGREED · HI-FP-U92-20260903-015

Canonical module maturity should follow M0 portfolio justification → M1 governance/roadmap → M2 development baseline → M3 capability implementation → M4 verification readiness → M5 adversarial/resilience verification → M6 release candidate → M7 staging/shadow/canary → M8 production → M9 managed self-evolution.

AGREED · HI-FP-U129B-20260905-001

Logistics Service is the reference pilot for reusable module self-knowledge, fresh-worker, coordination and reporting behavior; proven patterns should later be propagated to other canonical modules.

Context: Modules inherit the behavioral contract, not necessarily Logistics domain internals or byte-for-byte code.
Roadmap: BP-H10-REFERENCE-ROLLOUT

AGREED · HI-FP-U129B-20260905-002

The first substantial Logistics execution pool is fully manually gated; every prompt result is reviewed by the operator before the next prompt is released.

Context: The first pool calibrates quality, continuity and reporting before any automation widening.
Roadmap: BP-H10-POOL-CONTROL

AGREED · HI-FP-U129B-20260905-003

Later pools may use full-manual, checkpointed or automated child progression depending on prior-pool review, but every pool always ends at a human review gate.

Context: Automation level is selected by the operator and never widens itself.
Roadmap: BP-H10-POOL-CONTROL

AGREED · HI-FP-U129B-20260905-004

Machine closure of a child prompt inside an approved pool is conformance/progression evidence and must not be silently redefined as Blueprint ACCEPT.

Context: Preserve explicit human ACCEPT/RETURN/HOLD semantics while enabling future intra-pool automation.
Roadmap: BP-H10-POOL-CONTROL

AGREED · HI-FP-U129B-20260905-005

All modules must report prompt completion through one normalized semantic envelope so Blueprint and Inspector can validate them with shared tooling rather than interpreting different prose formats.

Context: Module internals may differ; completion evidence shape and semantics should be standardized.
Roadmap: BP-H10-UNIFIED-REPORTING

AGREED · HI-FP-U129B-20260905-006

The target execution model starts a fresh AI worker for each prompt and reconstructs continuity from repository-owned AGENTS, inventory/index, current authority, current prompt/contract/oracle and previous completion evidence.

Context: Conversation memory must not be required for safe continuation.
Roadmap: BP-H10-FRESH-WORKER

AGREED · HI-FP-U129B-20260905-007

After the first Logistics pool, perform an independent inventory/audit and compare worker claims against repository reality before selecting the next automation level or propagating the reference profile.

Context: Self-reporting alone is insufficient to decide whether automation is mature.
Roadmap: BP-H10-REFERENCE-ROLLOUT

AGREED · HI-FP-U129B-20260905-008

Temporary assistant/operator artifacts must have semantic unique non-overwriting names; cleanup permission is optional and the operator may retain archives for later analysis.

Context: Artifact identity should survive relocation outside its original tmp session folder.
Roadmap: BP-ASSISTANT-HANDOFF

AGREED · HI-FP-U129B-20260905-009

In Ukrainian the AI assistant uses feminine grammatical self-reference and addresses the human operator with masculine grammatical forms.

Context: Make the convention explicit in AGENTS-linked documentation so it is not repeatedly clarified.
Roadmap: BP-ASSISTANT-INTERACTION

AGREED · HI-FP-U131-20260906-001

Module inventory is intended as long-lived implementation memory: it must explain purpose, provenance, lifecycle, lineage, interfaces, dependencies, evidence, duplicates, roadmap role and retirement impact rather than acting as a file list.

Context: The current Inventory Program is the foundation and should be extended rather than replaced.
Roadmap: BP-H10-INVENTORY-VNEXT

AGREED · HI-FP-U131-20260906-002

Blueprint Control Plane evolution runs in parallel with the Logistics pilot and must reuse existing transparency, repository-knowledge freshness, safe-mutation and context-bundle primitives instead of creating competing frameworks.

Context: The Logistics pilot must not be blocked by broad control-plane modernization unless a concrete safety dependency exists.
Roadmap: BP-H10-CONTROL-PLANE

AGREED · HI-FP-U131-20260906-003

Critical current-state projections need explicit freshness and dependency contracts so a source change can deterministically identify which AGENTS, inventory, lineage, index, roadmap or context surfaces became stale.

Context: Fresh workers should start from a coherent snapshot with freshness proof, not from documents merely assumed current.
Roadmap: BP-H10-FRESHNESS

AGREED · HI-FP-U131-20260906-004

Significant implementation surfaces should eventually carry a short language-independent machine-readable metadata passport that identifies semantic role and points to durable inventory/lineage without embedding a full historical log in source code.

Context: Trivial helpers and generated/support surfaces should not create global metadata bureaucracy.
Roadmap: BP-H10-IMPLEMENTATION-METADATA

AGREED · HI-FP-U131-20260906-005

ForPrint should converge gradually on stable shared semantic IDs, aliases, contracts, events and statuses without a mass rename; old implementations may be mapped or migrated on touch while shared semantics become increasingly standardized.

Context: Semantic identity, implementation name and contract version are separate concepts.
Roadmap: BP-H10-SEMANTIC-CONVERGENCE

AGREED · HI-FP-U131-20260906-006

Before creating a significant potentially shared implementation, workers should query the existing Library semantic reference; missing or insufficient semantics should use registration/enrichment proposals and explicit provisional bindings without blocking ordinary local work.

Context: Cross-module breaking or high-risk interfaces may still require approval before integration.
Roadmap: BP-H10-SEMANTIC-CONVERGENCE

AGREED · HI-FP-U131-20260906-007

Project Inspector should use Library semantics, module inventory, implementation metadata and dependency relationships to filter candidates before deep source analysis and report whether matches are canonical, alias-based, provisional or inferred.

Context: The objective is to reduce AI guessing and focus human attention on ambiguous inferred cases.
Roadmap: BP-H10-INSPECTOR-CONFIDENCE

AGREED · HI-FP-U131-20260906-008

Operator reconciliation post_cf06_evening_handoff_reconciliation_v0_1: after CF-07/CF-08/CF-09, run one bounded internal Blueprint manual/shadow worker to validate the governed worker stack; after internal stability, Logistics remains the first external/module pilot; bootstrap must front-load Z-package intake, contradiction reconciliation and human-readable roadmap reporting; broad autonomy, release, push, merge and cross-repository writes remain blocked until later gates.

Context: The objective is to reduce AI guessing and focus human attention on ambiguous inferred cases.
Roadmap: BP-H10-INSPECTOR-CONFIDENCE

RECOVERED · HI-FP-ARCH-20260920-001

Workers may reason beyond their execution scope, but effective mutation authority must be machine-enforced and bounded by project policy, module policy, execution profile and the current work lease.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff; runtime guardrails must not depend on prompt memory.

RECOVERED · HI-FP-ARCH-20260920-002

Before introducing a significant architectural capability, Blueprint must audit existing and adjacent mechanisms and record REUSE, EXTEND, ADAPT, REPLACE or NEW; NEW requires explicit search evidence and rationale.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff; prevents parallel control planes and duplicate frameworks.

RECOVERED · HI-FP-ARCH-20260920-003

ForPrint should maintain distinct Portfolio/Planning, Implementation/Traceability and Execution/Runtime graph semantics so desired architecture, implementation evidence and actual execution facts are not collapsed into one ambiguous source.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff.

RECOVERED · HI-FP-ARCH-20260920-004

Publication is a separate authority boundary from execution completion: WORK_COMPLETED is not MERGED, and canonical Git promotion belongs to a governed publication path rather than Dispatcher or worker self-approval.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff.

RECOVERED · HI-FP-ARCH-20260920-005

Reusable capabilities should have lifecycle, lineage, version, release-manifest and consumer-review semantics; publication informs consumers but must not force every consumer to upgrade immediately.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff.

AGREED · HI-FP-ARCH-20260920-006

After portfolio knowledge is sufficiently saturated, Blueprint should maintain a dynamic planning/convergence capability that automatically recomputes dependencies, dependency matrices, affected chains, readiness/frontier and other derived planning projections when canonical roadmap knowledge changes.

Context: Recovered from 2026-09-20 NT-07 and explicitly reconfirmed by the operator on 2026-09-21.

AGREED · HI-FP-ARCH-20260920-007

Automatic roadmap convergence means automatic recomputation of derived planning knowledge, not automatic strategic reprioritization, work activation, ACCEPT, release, push or merge without separate authority.

Context: Explicit operator intent for the future portfolio planning engine; preserves planning/execution authority separation.

AGREED · HI-FP-ARCH-20260920-008

The portfolio planning system should expose machine-readable reports and human dashboards for dependencies, blockers, readiness, plan revisions, freshness, progress and portfolio effects, while keeping dashboards as projections rather than sources of truth.

Context: Operator requires automatic movement to be observable and reportable; metrics must preserve provenance and uncertainty.

AGREED · HI-FP-ARCH-20260920-009

Before implementing the portfolio convergence engine, the implementing assistant must re-audit the current Blueprint structure, reconcile existing mechanisms and enrich/refine the task so the solution extends current architecture instead of freezing assumptions from an older handoff.

Context: Explicit operator requirement on 2026-09-21.

RECOVERED · HI-FP-ARCH-20260920-010

Indexes, relationship graphs and human-readable architecture/dashboard views should remain complementary projections over canonical sources, with explicit invalidation and rebuild rules rather than competing sources of truth.

Context: Recovered from the 2026-09-20 Blueprint architecture handoff.

RECOVERED · HI-FP-ARCH-20260920-011

Planning desired state, actual execution history, execution lease control and desired-vs-actual reconciliation must remain separate responsibilities: Planner, Continuity/Event Store, Dispatcher and Reconciler must not silently take each other's authority.

Context: Recovered from NT-07 Portfolio Planning/Convergence architecture.

RECOVERED · HI-FP-ARCH-20260920-012

CPM, CCPM, PERT, Monte Carlo, sensitivity analysis, EVM, Lean/Last Planner, system dynamics and similar external methods are strategic reference mechanisms only until data maturity and deliberate Blueprint promotion justify implementation.

Context: Recovered from the far-horizon catalog; do not convert the whole catalog into backlog.

RECOVERED · HI-FP-ARCH-20260919-001

A durable long-running business Process Manager should not be created as a standalone module before auditing existing hosts; the first implementation should be hosted/incubated with its own contracts, state model, tests, adapters and explicit extraction readiness.

Context: Recovered from the 2026-09-19 architecture handoff; later portfolio planning selected Operations Control Registry as the working host.

RECOVERED · HI-FP-ARCH-20260919-002

AI in customer/order interpretation is a fallback/helper after deterministic rules, history, classifiers, customer context and clarification; unresolved ambiguity, lack of progress or contradictory evidence should drive escalation to a human.

Context: Recovered from the 2026-09-19 architecture handoff.

RECOVERED · HI-FP-ARCH-20260917-001

ForPrint planning must preserve four distinct mechanisms: Roadmap for committed executable work, Strategic Vector for current directional focus, Architecture/Improvement Horizon for non-executable candidates, and Review Obligations for subjects that must be revisited when triggers become due.

Context: Recovered from the 2026-09-17 evening handoff; these surfaces must not collapse into one planning authority.

RECOVERED · HI-FP-ARCH-20260917-002

Strategic Vector is a persistent coordination mechanism composed of finite versioned strategic epochs; each epoch has explicit completion or supersession criteria and does not override the executable roadmap.

Context: Recovered from the 2026-09-17 Strategic Vector lifecycle agreement.

RECOVERED · HI-FP-ARCH-20260917-003

Architecture/Improvement Horizon must permanently preserve promising ideas without granting roadmap or execution authority and must support WATCH, ASSESS, TRIAL, READY_FOR_DECISION, ADOPTED, REJECTED and SUPERSEDED states.

Context: Recovered from the 2026-09-17 handoff and later materialized in current global policy.

RECOVERED · HI-FP-ARCH-20260917-004

The Horizon requires mandatory reassessment at the first sufficiently mature coordinated multi-module operating point and may receive lighter reassessment at major wave boundaries.

Context: Recovered from the 2026-09-17 handoff; current Horizon global policy already carries this direction.

RECOVERED · HI-FP-ARCH-20260917-005

Review Obligations must be machine-readable, trigger-driven and evidence-based: a subject remains due until explicit completion criteria and a durable review disposition are satisfied; a reminder alone is insufficient.

Context: Recovered from the 2026-09-17 Review Obligations design seed and reconciled with planned CF-12.
Roadmap: CF-12

RECOVERED · HI-FP-ARCH-20260917-006

Review Obligation triggers should support time, milestone/wave, event, metric/health, dependency-change and strategic conditions, while dashboard/CRM views remain projections and Dispatcher must not self-authorize resulting governance decisions.

Context: Recovered from the 2026-09-17 design seed and current CF-12 direction.
Roadmap: CF-12

RECOVERED · HI-FP-ARCH-20260917-007

Fresh global-assistant bootstrap packages must be generated from live state rather than stale cached snapshots and expose roadmap cursor, lifecycle state, resume coordinates, strategic direction, human-readable current stage and stale-context failure behavior.

Context: Recovered from the 2026-09-17 next-session agreements.

RECOVERED · HI-FP-ARCH-20260917-008

Operator-facing status, bootstrap and dashboards must pair opaque machine IDs with human-readable stage or task meaning, short purpose, current state and next meaningful action while retaining IDs/sequences for correlation.

Context: Recovered from the 2026-09-17 bootstrap reporting rule.

RECOVERED · HI-FP-ARCH-20260915-001

The path to the first internal Blueprint worker should remain the minimum CF-08 → CF-09 → CF-10 sequence; later convenience and self-hardening work must not be inserted ahead of it unless a true dependency blocks launch.

Context: Recovered from the 2026-09-15 transition-to-autonomous-workers package and materialized in the current Strategic Vector/Control Foundation.
Roadmap: CF-08/CF-09/CF-10

RECOVERED · HI-FP-ARCH-20260915-002

After the first internal Blueprint worker becomes usable, bounded routine Blueprint self-hardening should increasingly move to that worker while the global/chat assistant shifts toward module audits, current-state evidence, target states, roadmaps, provenance, dependencies and portfolio planning.

Context: Recovered from the 2026-09-15 strategic transition package; this is the origin of the current parallel work split.

RECOVERED · HI-FP-ARCH-20260915-003

Broad autonomous module rollout must wait for a portfolio model covering module priority, dependency ordering, readiness, balancing/catch-up waves, forward-development waves and an explicitly authorized first coordinated worker wave.

Context: Recovered from the 2026-09-15 transition package and materialized primarily in CF-17/CF-19.
Roadmap: CF-17/CF-19

RECOVERED · HI-FP-ARCH-20260915-004

Early worker retries should be observation-first: looser but bounded, with hard runaway ceilings and detailed telemetry for failure fingerprints, repair actions, retries, restarts and resume coordinates; tighter production thresholds should be derived from measured evidence.

Context: Recovered from worker_retry_telemetry_ramp_v0_1 and materialized in CF-09/CF-10/CF-15.
Roadmap: CF-09/CF-10/CF-15

RECOVERED · HI-FP-ARCH-20260915-005

Established project functionality should progressively move behind registered capabilities/tools with describe/help, validation, authority gating, governed mutation and affected validation; protected direct writes should eventually fail

with actionable guidance to the required capability.

Context: Recovered from the 2026-09-15 Golden Path direction and materialized in CF-13/CF-18/CF-19.
Roadmap: CF-13/CF-18/CF-19

RECOVERED · HI-FP-ARCH-20260915-006

Internal worker self-hardening should begin with simple, observable and reversible tasks and increase complexity only after evidence; the first worker does not own global strategy, roadmap authority, broad rollout, release/push/merge or unrestricted cross-repository writes.

Context: Recovered from internal_worker_self_hardening_backlog_v0_1.
Roadmap: CF-10

RECOVERED · HI-FP-EVE-20260915-001

Evening, travel and weekend architecture discussions are first-class project work but their handoff package is non-executable: a later intake must audit current canonical state, classify each item, surface conflicts, map it to existing roadmap/policy and execute only currently due work.

Context: Recovered from the separate 2026-09-15 evening handoff package.

RECOVERED · HI-FP-EVE-20260915-002

Critical startup rules for a fresh coordinating assistant must be explicit and immediately visible rather than buried, including current authority, roadmap/workfront/lifecycle, handoff/intake mode, conflict handling, unresolved questions, provenance and human-readable reporting.

Context: Recovered from bootstrap_strengthening_requirements.md.

RECOVERED · HI-FP-EVE-20260915-003

Operator-facing roadmap/status reporting should explain finished work, current stage, why it matters and the next meaningful steps in plain language; machine IDs are secondary traceability metadata rather than the primary explanation.

Context: Recovered from human_readable_operator_reporting.md.

RECOVERED · HI-FP-EVE-20260915-004

Temporary tmp artifacts are disposable; required execution continuation must live in durable attempt/checkpoint/resume state with failure reason, validation state, retry lineage and final evidence.

Context: Recovered from runtime_execution_and_verification_requirements.md and later materialized in CF-09/runtime contracts.
Roadmap: CF-09

RECOVERED · HI-FP-EVE-20260915-005

Verification should be tiered so small changes can use local/related/module checks while project/full and nightly/deep checks run when warranted; heavy full validation must not be the default after every bounded change.

Context: Recovered from runtime_execution_and_verification_requirements.md and materialized directionally in CF-14.
Roadmap: CF-14

RECOVERED · HI-FP-EVE-20260915-006

Before autonomous workers are trusted in older modules, each module needs a repository-grounded deep audit, stale/conflicting documentation and duplicate/obsolete implementation review, and a human-built knowledge/index baseline.

Context: Recovered from the 2026-09-15 evening roadmap intake and module rollout strategy.

RECOVERED · HI-FP-EVE-20260915-007

Dispatcher should eventually have a small dedicated local Telegram operator-control adapter for status, diagnostics, alerts and bounded remote actions; it is separate from the full customer-facing Telegram module.

Context: Recovered from dispatcher_local_telegram_bot_requirements.md and currently bound to CF-10/CF-15/CF-18 rather than a new roadmap step.
Roadmap: CF-10/CF-15/CF-18

RECOVERED · HI-FP-EVE-20260915-008

Privileged remote Dispatcher actions must use explicit audited elevation or break-glass semantics instead of permanent unrestricted authority, and Dispatcher health should be independently supervised with bounded restart/escalation.

Context: Recovered from dispatcher_local_telegram_bot_requirements.md.
Roadmap: CF-10/CF-15/CF-18

RECOVERED · HI-FP-ARCH-20260912-001

Important recurring or authority-sensitive work should execute through versioned Governed Procedure Graphs with conditional steps, hard/observed gates, required contracts, evidence and a Run Manifest rather than relying on prompt memory.

Context: Recovered from the 2026-09-12 architecture planning session and now represented in CF-07.
Roadmap: CF-07

RECOVERED · HI-FP-ARCH-20260912-002

An evening or strategic human intent must not be copied directly into roadmap implementation; it must first search existing capabilities and roadmap items, resolve ownership/dependencies, adapt to real architecture and produce a semantic diff with explicit operator gates for narrowing, removal, breaking changes or major ownership change.

Context: Recovered from the 2026-09-12 morning-reconciliation design and strengthened later by the Reuse-First / Pre-Implementation Reconciliation Gate.

RECOVERED · HI-FP-ARCH-20260912-003

Long-lived project memory has three distinct layers: canonical project state, a Decision/Rationale Ledger explaining why and alternatives, and immutable raw conversation evidence referenced by stable identity/hash; raw dialogue is evidence, not authority.

Context: Recovered from the 2026-09-12 architecture planning session and mapped to CF-11.
Roadmap: CF-11

RECOVERED · HI-FP-ARCH-20260912-004

Human Portfolio must be generated from canonical machine state plus rationale/history rather than maintained as an independently edited copy, and should support Current-to-Target, Full Historical and Decision/Change Review views.

Context: Recovered from the 2026-09-12 portfolio model and mapped to CF-18.
Roadmap: CF-18

RECOVERED · HI-FP-ARCH-20260912-005

Cross-module workers may OBSERVE, QUERY and PROPOSE but must not silently FOREIGN MUTATE; improvement proposals are collected centrally, deduplicated and reconciled with roadmap/ownership before they can become project work.

Context: Recovered from the 2026-09-12 cross-module model and mapped to CF-16.
Roadmap: CF-16

RECOVERED · HI-FP-ARCH-20260912-006

A missing provider capability should not automatically stop all consumer work: safe bounded temporary compatibility stubs may create explicit Dependency Debt with provider, required capability, target step and replacement trigger.

Context: Recovered from the 2026-09-12 wave/dependency discussion and mapped to CF-16.
Roadmap: CF-16

RECOVERED · HI-FP-ARCH-20260912-007

Project Health should be one canonical machine snapshot covering module health, knowledge freshness, procedure conformance, graph drift, tests, dependency debt, semantic review obligations, publication state, wave balance and worker/context health; dashboards and alerts are projections over it.

Context: Recovered from the 2026-09-12 control-plane design and mapped to CF-15.
Roadmap: CF-15

RECOVERED · HI-FP-ARCH-20260912-008

Validation should use deterministic local, affected, core and full tiers, with affected tests derived from dependency/validation mappings and Full Check Debt driven by critical surfaces, fan-out, elapsed time, contracts and lifecycle/wave/publication context.

Context: Recovered from the 2026-09-12 test/status model and mapped to CF-14.
Roadmap: CF-14

RECOVERED · HI-FP-ARCH-20260912-009

Every module should become deterministically machine-discoverable through explicit purpose, owner, capabilities, entrypoints, commands, tests, contracts, procedures, dependencies, deprecated surfaces, roadmap/current work and authority; new modules follow the golden path while old modules improve incrementally on touch.

Context: Recovered from the 2026-09-12 AI-legibility discussion and mapped to CF-13.
Roadmap: CF-13

RECOVERED · HI-FP-ARCH-20260912-010

Blueprint AI autonomy must progress by capability rather than one global autonomous flag: Shadow, Copilot, Bounded Operator, Autonomous Routine Coordinator and only later High Autonomy by Capability, with slower promotion for larger blast radius.

Context: Recovered from the 2026-09-12 Blueprint AI Coordinator model and mapped to CF-19/CF-20.
Roadmap: CF-19/CF-20

RECOVERED · HI-FP-ARCH-20260912-011

A Blueprint AI actor must never be able to change the guardrail that constrains it, validate that same change itself and declare PASS; high-impact control-plane guardrails require separation of duties or independent/operator validation.

Context: Recovered from the 2026-09-12 Blueprint AI safety discussion and mapped to CF-19/CF-20.
Roadmap: CF-19/CF-20

RECOVERED · HI-FP-ARCH-20260912-012

Roadmap is desired plan/spec while lifecycle/event evidence is actual execution truth; current/previous/next/complete execution status must be deterministically derived rather than manually duplicated in both surfaces.

Context: Recovered from roadmap_state_reconciliation_design_v0_1 and materialized as CF-02.
Roadmap: CF-02

RECOVERED · HI-FP-ARCH-20260912-013

A Work Front is durable project evidence of what and why work was assigned, while generated assistant packs are disposable projections; repeated executions of one Work Front have separate execution-attempt records with profile, pack/source fingerprints, budget/tool use, result, validator and retry reason.

Context: Recovered from the 2026-09-12 handoff/work-front discussion and mapped to CF-05.
Roadmap: CF-05

RECOVERED · HI-FP-ARCH-20260912-014

Execution profiles must separate reasoning depth, authority and budget, and effective authority follows Project Constitution → Module Policy → Execution Profile → Work Front, where lower layers may narrow but cannot silently widen higher-level authority.

Context: Recovered from the 2026-09-12 execution-profile discussion and mapped to CF-03/CF-06.
Roadmap: CF-03/CF-06

RECOVERED · HI-FP-ARCH-20260912-015

Publication must be a separate governed pipeline from development completion; commits operate on exact allowed path sets rather than broad git add -A on a dirty tree, and push/PR/merge authority increases only through explicit publication gates.

Context: Recovered from the 2026-09-12 Git/publication discussion and mapped to CF-18.
Roadmap: CF-18

RECOVERED · HI-FP-FORENSICS-20260908-001

Blueprint needs a repeatable portfolio-forensics capability that compares each module against the full ForPrint course from mission and policy through target architecture, module role, dependencies and verified actual state to alignment/gap, including contradictions, obsolete implementations and duplicate semantics.

Context: Recovered from the unique but partial 08.09 dialogue; this is the origin of the machine-readable module X-ray workflow.

RECOVERED · HI-FP-FORENSICS-20260908-002

Module status must not collapse implementation state, roadmap/planning state, execution permission and dispatch eligibility into one label; semantic freshness is another distinct dimension that may fail even when structural tests pass.

Context: Recovered from B10/Logistics forensic findings on 08.09 and compatible with later current-state and CF-02 reconciliation semantics.
Roadmap: CF-02

RECOVERED · HI-FP-FORENSICS-20260908-003

A final module disposition such as KEEP, RECONCILE, PAUSE, DEPRECATE, REMOVE or ADVANCE must use local repository evidence together with the current project course, ownership, dependencies and relevant cross-repository/runtime evidence; local health alone is insufficient.

Context: Recovered from the 08.09 portfolio-forensics workflow.

RECOVERED · HI-FP-FORENSICS-20260908-004

Portfolio forensic coverage and unresolved/unnormalizable findings must remain machine-visible: missing evidence becomes an explicit gap or operator question rather than being guessed away, and any coverage percentage must preserve its denominator or calculation basis.

Context: Recovered from the 08.09 coverage-dashboard and unresolved-registry requirements.

RECOVERED · HI-FP-FORENSICS-20260908-005

Canonical authority/source lineage must be reconciled before generated guides, diagrams, status reports or other projections are regenerated; a derived PASS must not conceal stale current-step or broken lineage assumptions.

Context: Recovered from the 08.09 B10 semantic-review findings.

RECOVERED · HI-FP-FORENSICS-20260908-006

Forensic collectors must fail closed on repository identity, report nonzero check return codes as FAIL rather than successful capture, and avoid repeating passes that add no new evidence class.

Context: Recovered from the 08.09 Logistics L2 collector correction and Blueprint collector mislaunch.

RECOVERED · HI-FP-FORENSICS-20260908-007

Temporary forensic tooling may be disposable, but its logic and workflow must be reconstructible from durable project instructions so a replacement assistant can continue collector → evidence package → analysis → next collector without chat memory.

Context: Recovered from the 08.09 continuity/tooling requirements.

RECOVERED · HI-FP-FORENSICS-20260908-008

When domain implementation is healthy but self-knowledge, prompt lifecycle, freshness or control-plane projections are stale, the preferred response is to preserve the healthy business code and reconcile the control plane rather than rewrite the module merely to match stale metadata.

Context: Recovered from the historical Logistics verdict KEEP_MODULE_AND_RECONCILE_CONTROL_PLANE.

RECOVERED · HI-FP-DTFB-20260905-001

When multiple historical memory/Human Intent packages describe the same dialogue under different ID schemes, Blueprint should choose the stronger normalized source, preserve older material as residual evidence and merge only semantic deltas rather than duplicate equivalent intents.

Context: Recovered from the 05.09 v1.0/v1.1 reconciliation decision; complements current roadmap-enrichment source intake.

RECOVERED · HI-FP-DTFB-20260905-002

Governance closeout is not complete merely because a code or document mutation exists: required rules and descriptions must propagate to the intended project/module instructions, remain discoverable in indexes, and pass relevant health/conformance checks.

Context: Recovered from the 05.09 self-cleaning/closeout discussion.

RECOVERED · HI-FP-DTFB-20260905-003

Blueprint publishes released prompts only through Blueprint-owned surfaces; module-side intake pulls/reads them into the module execution flow and must not use the listener as authority to write foreign Blueprint state.

Context: Recovered from the 05.09 canonical prompt-pull architecture and preserved as repository-boundary provenance.

RECOVERED · HI-FP-DTFB-20260905-004

One-shot prompt release authorization must return to fail-closed behavior after the bounded release transaction and must not silently widen into standing release authority.

Context: Recovered from the historical generic-health release hardening; current implementation must be audited if this path is modified.

RECOVERED · HI-FP-DTFB-20260905-005

Post-release prompt prepared-buffer membership and roadmap Acceptance Oracle binding are distinct concepts: releasing a prompt may remove it from prepared drafts but must not erase the durable acceptance contract for its roadmap work.

Context: Recovered from the 05.09 Acceptance Oracle edge case.

RECOVERED · HI-FP-DTFB-20260905-006

Current governance/evidence documents should retain durable inbound semantic references; if a current document becomes orphaned, repair the canonical reference graph rather than weakening the semantic validator to hide the problem.

Context: Recovered from the 05.09 GENUINE_NO_INBOUND_CURRENT_DOCUMENTS investigation.

RECOVERED · HI-FP-DTFB-20260905-007

Self-inventory/system-context and portfolio/roadmap snapshots are portable navigation and evidence projections for fresh assistants, not independent current authority and not a substitute for live authority/freshness validation.

Context: Recovered from the 05.09 snapshot workflow.

RECOVERED · HI-FP-GENESIS-20260714-001

ForPrint System Blueprint exists to preserve architecture and coordination truth across independently evolving modules; it is not the owner of module domain logic, while Project Inspector independently verifies implementation against declared architecture and evidence.

Context: Recovered from the 14.07 architecture genesis; later portfolio work matured this Blueprint/Inspector separation.

RECOVERED · HI-FP-GENESIS-20260714-002

The ecosystem north star is maximum practical automation of routine print-shop work, while later governance must bound autonomy by authority, risk, validation and human exception/strategic control rather than treating automation as unrestricted execution.

Context: Recovered from the owner's 14.07 foundational automation objective and reconciled with later AI/autonomy governance.

RECOVERED · HI-FP-GENESIS-20260714-003

Architecture changes should be versioned and machine-discoverable, and meaningful Blueprint changes should trigger review of affected modules and regenerated guidance rather than relying on assistants to remember architecture changes from chat.

Context: Recovered from the 14.07 manifest/ADR/affected-module/guidance discussion; later reflected in machine discoverability and change-impact work.

Roadmap: CF-13/CF-14

RECOVERED · HI-FP-GENESIS-20260714-004

Shared assistant governance should standardize interfaces, required evidence and common rules globally without forcing every module to use one identical internal script or duplicating already established coordination paths.

Context: Recovered from 14.07 assistant-coordination agreements and consistent with later reuse-first governance.

RECOVERED · HI-FP-GENESIS-20260714-005

Agreed work must enter durable repository-owned roadmap/queue/completion/reporting surfaces instead of remaining only in conversation memory, and executed prompt drafts should transition out of active draft state rather than accumulate indefinitely.

Context: Recovered from the 14.07 durable queue, roadmap, completion packet, reporting and prompt-lifecycle discussion.

RECOVERED · HI-FP-GOV-20260827-001

ForPrint governance must evolve forward: when an older rule obstructs an accepted newer architecture, update, supersede or deprecate the old rule explicitly and preserve authority lineage rather than forcing new design to imitate obsolete constraints.

Context: Recovered from the 27.08 owner decision; this is controlled governance evolution, not permission to ignore current authority.

RECOVERED · HI-FP-GOV-20260827-002

Durable project history should preserve significant decisions and rationale while avoiding unnecessary directory and intermediate-revision proliferation; history must remain navigable, with current versus older authority obvious.

Context: Recovered from the 27.08 project-memory and repository-structure discussion; complements later CF-11 provenance work.

Roadmap: CF-11

RECOVERED · HI-FP-GOV-20260827-003

Long or complex assistant scripts should be delivered as files for execution, while terminal instructions should stay short and practical; temporary script identity must remain unambiguous to avoid running stale revisions.

Context: Recovered from the 27.08 operator workflow and repeated stale-script incidents.

RECOVERED · HI-FP-GOV-20260827-004

Unrelated dirty worktree state must not become a global blocker for Blueprint automation: execution compatibility is evaluated against bounded required inputs, authority, baseline identity, owned mutation scope and conflict/freshness guards; true parallel execution must use isolated workspaces rather than concurrent writes into one shared dirty checkout.

Context: Recovered from the 27.08 parallel-work doctrine and reconciled with later execution-lane rules.

Roadmap: CF-10

RECOVERED · HI-FP-GOV-20260827-005

Mutation tooling must verify artifact/version identity and expected structure before changing repository state, fail closed on stale or ambiguous inputs, and preserve unrelated work on both success and rollback.

Context: Recovered from the 27.08 stale-revision and fail-closed mutation incidents; later tooling strengthens this with hashes/preimages/protected surfaces.

RECOVERED · HI-FP-PREFLIGHT-20260831-001

Deep repository understanding should combine bounded machine inventory with bounded semantic review and persisted coordinator synthesis; machine/external findings remain evidence until authority, conflicts and actionable references are reconciled.

Context: Recovered from the 31.08 Blueprint self-inventory methodology and now aligned with Roadmap Enrichment / Knowledge Saturation.

RECOVERED · HI-FP-PREFLIGHT-20260831-002

Generated capability/knowledge indexes should be rebuilt only after contradictory, duplicate or stale semantic sources are reconciled; a cleaner generated projection must not hide unresolved source-authority conflicts.

Context: Recovered from the 31.08 cleanup/index ordering decision.

RECOVERED · HI-FP-PREFLIGHT-20260831-003

Portfolio planning should support machine/deep-navigation, balanced technical/operator and extended human-readable representations from shared underlying state, while preserving roadmap binding, Human Intent provenance and visibly non-authoritative synthetic future planning.

Context: Recovered from the 31.08 three-view portfolio model; later portfolio/Human Intent surfaces mature this direction.

RECOVERED · HI-FP-PREFLIGHT-20260831-004

Legacy print-file and folder naming may encode real operational semantics; migration must preserve retrieval and captured meaning while moving toward canonical metadata, and historical naming conventions are migration evidence rather than automatic final standards.

Context: Recovered from the 31.08 legacy file-semantics discussion affecting Prepress, Calculator and Library.

RECOVERED · HI-FP-PREFLIGHT-20260831-005

A cross-module standard is not adopted merely because its document exists: affected modules need explicit roadmap/prompt adoption work and readiness/conformance evidence before shared activation.

Context: Recovered from the 31.08 shared-standard rollout discussion and consistent with later dependency/adoption governance.

RECOVERED · HI-FP-ORIGIN-20260805-001

Repository Knowledge must represent evidence-backed understanding rather than scaffolding or file coverage: material claims should preserve confidence/evidence classification and exact repository baseline provenance, and unknown/conflicting knowledge must stay explicit.

Context: Recovered from the 05.08 rejection of template-only knowledge and the historical index-vs-understanding gap.

RECOVERED · HI-FP-ORIGIN-20260805-002

A public command named check or status-check must be read-only; any operation that changes tracked project state, authority, release state or durable coordination data must use explicit mutating command semantics rather than hiding writes behind a check.

Context: Recovered from the 05.08 Make/command-architecture correction.

RECOVERED · HI-FP-ORIGIN-20260805-003

Bounded external semantic analysis must remain tied to the exact originating request/workflow and content checksum, and only a validated response is merged back into the owning project workflow; external analysis is evidence, not free-standing authority.

Context: Recovered from the 05.08 Blueprint self-audit external-assessment roundtrip.

RECOVERED · HI-FP-ORIGIN-20260805-004

Validation success, readiness and execution or rollout authorization are distinct states: a fully green canonical gate must not silently authorize a pilot, release, external rollout or wider autonomy without the separate authority decision required by current policy.

Context: Recovered from the 05.08 mutation-builder/pilot boundary and later generalized by Control Foundation authority gates.

RECOVERED · HI-FP-ORIGIN-20260614-001

Portfolio planning must remain dependency-constrained: Blueprint keeps whole-system and per-module plans aligned, prevents a module from outrunning unavailable upstream contracts or capabilities, and maintains meaningful forward direction without relying on a fixed numeric horizon.

Context: Recovered from the early source-label 14.06 coordination model and reconciled with later full-horizon roadmaps/dependency intelligence.

RECOVERED · HI-FP-ORIGIN-20260614-002

If no eligible next workfront exists, assistants must not invent one from chat context: resolve current authority and use roadmap, prepared/draft work and dependency evidence to propose or materialize the documented next step through the governed lifecycle.

Context: Recovered from the early prompt/roadmap resolver discipline.

RECOVERED · HI-FP-ORIGIN-20260614-003

Important reports must be semantically consumed rather than merely generated: each meaningful report should feed a decision, gap, follow-up action, review obligation or explicit no-action disposition so reporting does not become busywork.

Context: Recovered from the owner's early repository self-analysis/report-consumption correction.

RECOVERED · HI-FP-ORIGIN-20260614-004

Scripts and machine-operable project artifacts should expose machine-readable responsibility metadata such as purpose, dependencies and outputs/effects, and repository audit should compare declared metadata with observed behavior, flag drift and support reviewed batch reconciliation instead of constant manual header editing.

Context: Recovered from the early machine-readable script metadata and metadata-vs-code drift design; implementation must reuse current discoverability/Inspector mechanisms where possible.

Roadmap: CF-13

RECOVERED · HI-FP-ORIGIN-20260614-005

Temporary catalogs, fixtures or synthetic dependency data may be used only when explicitly bounded and non-authoritative; they must never masquerade as canonical domain truth or silently replace the real owning module's data contract.

Context: Reconciles the early no-virtual-data doctrine with later accepted bounded temporary/local catalogs used for development and testing.

RECOVERED · HI-FP-CLOSEDLOOP-20260805C-001

Module completion must become machine-discoverable through repository-owned coordination evidence and the normal Blueprint intake command chain; safe progression must not depend on someone notifying the Blueprint assistant in chat.

Context: Recovered from the 05.08 continuation completion-intake redesign and later matured into completion packet/outbox/intake semantics.

RECOVERED · HI-FP-CLOSEDLOOP-20260805C-002

Prompt issuance and completion reporting should form one structured exchange contract: prompt requirements, produced artifacts, commands, task-level evidence, final validation state and roadmap bindings must be recognizable to shared tooling.

Context: Recovered from the owner-defined prompt/report contract in the 05.08 continuation; later normalized completion-envelope work extends it.

RECOVERED · HI-FP-CLOSEDLOOP-20260805C-003

Roadmap depth and prepared-work buffers are health and planning-risk signals, not execution or acceptance authority; exact historical numeric thresholds may be superseded by current full-horizon planning and should not become hidden hard stops.

Context: Reconciles the historical 5-8 roadmap steps / 2 draft prompts rule with current portfolio planning.

RECOVERED · HI-FP-CLOSEDLOOP-20260805C-004

Accepted or closed prompts should leave actionable current/approved surfaces and remain available through completed/archive history and indexes, so current prompt views represent executable work rather than accumulated history.

Context: Recovered from the v0.4 prompt archive lifecycle.

RECOVERED · HI-FP-CLOSEDLOOP-20260805C-005

Parallel development should use attributable Git branches, worktrees or explicitly isolated execution workspaces instead of routinely synchronizing temporary full repository copies; reconciliation should return work to canonical Git state.

Context: Recovered from the owner's 05.08 continuation decision after repeated temp-repository synchronization overhead.
Roadmap: CF-10

RECOVERED · HI-FP-HARDENING-20260817-001

When a technical task is already explicitly bounded, finish it or explicitly stop/return it before repeatedly redesigning the roadmap around it; roadmap reconciliation should occur deliberately at meaningful task boundaries.

Context: Recovered from the 17.08 v0.4.1 hardening discussion.

RECOVERED · HI-FP-HARDENING-20260817-002

Roadmap planning should be able to represent a stable main task with meaningful subtasks so implementation depth and partial completion remain visible, while supporting a compact main-task view and a detailed task/subtask view from the same canonical state.

Context: Recovered from the 17.08 roadmap-granularity decision; current schema support requires reuse audit before implementation.

RECOVERED · HI-FP-HARDENING-20260817-003

Superseded tests, scripts and workflow mechanisms should not permanently block the current platform merely to preserve legacy compatibility; after replacement they may be deprecated, isolated or archived for historical access under controlled governance.

Context: Recovered from the 17.08 predictability/stability/transparency and legacy-retirement decisions; complements later self-cleaning governance.

RECOVERED · HI-FP-HARDENING-20260817-004

The operator may express business intent through practical examples, while Blueprint/assistant should translate that intent into technically correct, modular and reusable machine-oriented prompts or specifications rather than copy the examples literally.

Context: Recovered from the 17.08 prompt-design philosophy.

RECOVERED · HI-FP-HARDENING-20260817-005

A zero-context bootstrap or handoff must preserve the current position, exact next bounded work, remaining path and final objective so a replacement assistant can reconstruct the whole development line rather than receiving only the next command.

Context: Recovered from the late 17.08 continuity requirement and later matured into structured bootstrap/Handoff mechanisms.
Roadmap: CF-08

RECOVERED · HI-FP-CF02-ORIGIN-20260907-001

When governance or automation introduces new or changed operational command surfaces, persistent generators or standards, the same bounded change must register their execution dependencies, derivation lineage and canonical document/index discoverability rather than leaving hidden operational surfaces.

Context: Recovered from CF-02 candidate v0.1 findings R01-R03; later current checks indicate these mechanisms were materially developed.
Roadmap: CF-02

RECOVERED · HI-FP-CF02-ORIGIN-20260907-002

Future Control Foundation work may be planned while remaining unbound, but lifecycle or work IDs must not be silently pre-assigned beyond the currently authorized scope; identity binding is itself an authority-bearing transition.

Context: Recovered from CF-02 candidate v0.1 finding R04.
Roadmap: CF-02

RECOVERED · HI-FP-CF02-ORIGIN-20260907-003

A newly introduced persistent projection requires explicit bootstrap ordering: register/apply its producer and dependencies, materialize the first projection, then run read-only freshness or synchronization checks and broader validation gates.

Context: Recovered from CF-02 candidate v0.1 finding R05.
Roadmap: CF-02

RECOVERED · HI-FP-CF02-ORIGIN-20260907-004

Focused Control Plane tests must use explicit state/projection inputs or controlled fixtures and must not silently depend on ambient repository-root generated projections; production/runtime checks may still fail closed on stale or missing projections.

Context: Recovered from CF-02 candidate v0.1 finding R06.
Roadmap: CF-02

RECOVERED · HI-FP-CF02-ORIGIN-20260907-005

Legacy execution-state migration must bind to exact immutable pre-migration evidence such as source roadmap identity, accepted step IDs and evidence references; a coarse ordinal boundary alone is insufficient provenance.

Context: Recovered from CF-02 candidate v0.1 finding R07.
Roadmap: CF-02

RECOVERED · HI-FP-RUNTIME-ORIGIN-20260909-001

Automation may progress prompt-to-worker-to-report-to-validation-to-next-work inside an explicitly approved bounded work pool, but completion of a major pool, wave or phase requires explicit human/operator acceptance before authority widens to the next pool.

Context: Recovered from the 09.09 Logistics pilot operating model and aligned with later Operator Approval/Work Front/Strategic Transition boundaries.

RECOVERED · HI-FP-RUNTIME-ORIGIN-20260909-002

Blueprint should not routinely repair or normalize another module repository's Git state; module-local workers may prepare the bounded local baseline required for their task, but commit, push or merge authority must come from current higher policy, Execution Profile, Work Front or explicit approval rather than from module locality itself.

Context: Reconciles the 09.09 module-local preflight decision with later governed publication authority.

RECOVERED · HI-FP-RUNTIME-ORIGIN-20260909-003

Project startup must have one current canonical authority path, while stable mission/policy is kept separate from historical continuity evidence; historical context is optional evidence rather than default authority, and generated context should be selected by launch mode and task scope.

Context: Recovered from the 09.09 startup-normalization line and reconciled with later PROJECT_ONBOARD/TASK_EXECUTION Handoff v2 modes.
Roadmap: CF-08

RECOVERED · HI-FP-INFRA-ORIGIN-20260910-001

Project mutations should use a project-aware transactional mutation mechanism that validates candidate changes, may repair only bounded mechanically safe defects such as authorized syntax/lint issues, reruns required gates, records evidence and rolls back or fails closed on unsafe, ambiguous or gate-breaking results.

Context: Recovered from the 09.10 Mutation Compiler line; later current CF apply scripts visibly reuse scripts.coordination.mutation_compiler.

RECOVERED · HI-FP-INFRA-ORIGIN-20260910-002

Executable scripts, functions and Make targets must not depend on accidental undeclared files, caches, environment residue or ordering; prerequisites must be declared and validated or produced by an explicit supported generator, and critical paths should be testable in clean or sterile environments.

Context: Recovered from the 09.10 hidden-dependency/clean-room/execution-DAG hardening line.

RECOVERED · HI-FP-INFRA-ORIGIN-20260910-003

Long-running ForPrint commands and assistant-supplied scripts must expose visible phase/progress or heartbeat output while preserving full durable evidence; operator progress is an observability requirement and does not widen mutation, cleanup, publication or execution authority.

Context: Recovered from the 09.10 progress-visibility decision and later materialized as a project-wide execution progress contract.

RECOVERED · HI-FP-INFRA-ORIGIN-20260910-004

Generated knowledge, indexes, mirrors and runtime caches must be faithful and reproducible from declared authoritative inputs and derivation dependencies; drift or order/cache-dependent rebuild differences are defects to detect rather than hidden normal behavior.

Context: Recovered from the 09.10 faithful-mirror and cache-determinism hardening work.

RECOVERED · HI-FP-RECOVERY-ORIGIN-20260912-001

When a mutating operation fails after canonical state may already have advanced, recovery must not blindly rerun the original mutator. It must inspect the durable current state and fingerprints, identify the exact completed boundary, and repair only the failed downstream lifecycle/evidence/projection layer without repeating already-applied canonical source mutation.

Context: Recovered from 12.09 Control Foundation failures/recovery; CF-01 checkpoint-cardinality recovery is the concrete historical example.

RECOVERED · HI-FP-DIAGNOSTIC-TRUTH-20260914-001

Validation and diagnostic reports must be bound to the source or materialized snapshot they actually describe and must not be treated as current evidence after that provenance becomes stale. Before modifying source/tests, distinguish current-working-tree truth from isolated or non-mutating public-check materialization/report truth; if the current tree is green and the report/snapshot is stale, repair the isolation/materialization/report-freshness layer instead.

Context: Recovered from the 14.09 controlled-failure hash-pin/stale full-log diagnosis.

RECOVERED · HI-FP-ARTIFACT-INTEGRITY-20260914-001

A generated script or downloadable artifact must not be executed or accepted as complete until its delivery identity and completeness are verified. Truncated or ambiguous delivery requires regeneration under a distinct artifact identity and verification through checksum, byte/line counts, terminal marker or equivalent completeness evidence.

Context: Recovered from repeated truncated generated-script deliveries in the 14.09 continuation.

RECOVERED · HI-FP-OPERATOR-UX-ORIGIN-20260917-001

The Blueprint Makefile is an operator-facing command map, not merely a build file: public targets should be grouped and explain their purpose, safety or mutation semantics, required/default inputs, scope and result, while legacy, compatibility and alternate-purpose targets are explicitly classified without silently changing command behavior.

Context: Recovered from direct owner requirements in the 17.09 Control Foundation continuation; later materialized in the current Makefile operator guide.

RECOVERED · HI-FP-PROJECT-CLOSURE-ORIGIN-20260917-001

Focused or local subsystem readiness does not by itself authorize project closure. A full project boundary must still validate relevant cross-module dependencies, generated-state freshness and whole-project gates; if that boundary is red, lifecycle closure, worker dispatch and activation of the next automation phase remain blocked and recovery starts from the actual failed boundary.

Context: Recovered from CF-09 Dispatcher: 9/9 local readiness still failed full project closure on stale generator inventory.
Roadmap: CF-09

RECOVERED · HI-FP-PACKAGING-SCOPE-20260921-001

Assistant, bootstrap and living-knowledge packaging must stay within the explicitly authorized packaging/knowledge scope and must not silently redesign, reorder, reinterpret or mutate roadmap state merely to make a package self-contained; roadmap change requires its own explicit planning/mutation authority.

Context: Recovered from the 21.09 owner correction that instruction/package work had drifted into roadmap reconstruction.

RECOVERED · HI-FP-PACKAGE-IDENTITY-20260921-001

Generated assistant/context/handoff packages must expose and validate their semantic identity: package type, target module or explicit portfolio scope, purpose and intended use. Archive/hash validity alone is insufficient; unexpected implicit module routing or a mismatch between requested scope and emitted manifest/filename must fail closed or remain explicitly unresolved.

Context: Recovered from the wrong-package observation and the unexpected forprint_library project-context archive during Blueprint CF-10 preparation.

RECOVERED · HI-FP-DURABLE-PACKAGING-20260921-001

Once assistant/bootstrap package behavior or reusable living-knowledge packaging has been validated, its canonical logic, contract and reusable project-facing artifacts should live durably in the repository with provenance rather than existing only as sandbox/tmp output; publication of that knowledge still grants no execution, dispatch, release, commit or roadmap authority.

Context: Recovered from the 21.09 requirement to promote validated assistant/bootstrap packaging into durable project knowledge.

AGREED · HI-FP-CONTEXT-DEFAULT-20260922-001

When `make assistant-context-pack` is run from the Blueprint repository without an explicit `MODULE=...`, its default module scope is `forprint\_system\_blueprint`; an explicit command-line MODULE selects another canonical module. A global/default Makefile MODULE value must not silently route the package to an unrelated module.

Context: Owner decision on 2026-09-22 after observing an unexpected forprint_library context archive during Blueprint work.

AGREED · HI-FP-CONTEXT-PARALLEL-WORKSTREAM-20260922-001

The Blueprint assistant context package should carry a bounded, refreshable snapshot of active parallel workstreams so a fresh assistant can quickly understand the current engineering position, completed checkpoints, active write-set, near-term horizon, blockers and safe resume rules without relying on private chat memory. The snapshot is navigation/current-work context only and grants no execution or lifecycle authority.

Context: Operator handoff decision on 2026-09-22 while CF-10 Internal Worker Engineering and Roadmap Enrichment were being worked in parallel.

AGREED · HI-FP-MAKEFILE-FUNCTIONAL-MAP-20260922-001

ForPrint Makefile is an operator-facing functional map of the project, not merely a command launcher. Supported functionality that exposes an operator workflow must be represented in the appropriate Makefile section; newly implemented or newly discovered supported workflows trigger Makefile coverage review, and stale targets for functionality that no longer exists must be removed or explicitly reclassified as part of bounded task closeout.

Context: Owner decision 2026-09-22, reinforced by historical Make transparency/control-surface and operational-command discoverability lineage.

RECOVERED · HI-FP-LIBRARY-STRUCTURE-20260705-001

ForPrint repositories expected to accumulate many scripts or files should group new content into thematic subdirectories, generally keeping one level of nesting unless deeper structure has clear value; stable legacy layout should not be mass-moved merely for cosmetic consistency.

Context: Recovered from direct owner Library repository-structure rules in visible turns 89 and 93; current general architecture-hygiene material covers related placement concerns but does not preserve this one-level/thematic rule explicitly.

RECOVERED · HI-FP-REPORT-DEDUP-20260705-001

Operator-facing validation reports should consolidate semantically identical repeated checks or evidence rows, while retaining separate entries when they prove genuinely different evidence classes or semantics.

Context: Recovered from the owner reporting correction in visible turn 177; deduplication must not collapse distinct evidence classes.

RECOVERED · HI-FP-LIBRARY-ADMIN-20260516-001

Routine changes to Library-owned canonical references, rules and versions should be manageable through a strong human-friendly governed administration surface; adding a material, currency or similar reference should not require editing server code.

Context: Recovered from direct owner turn 1. Current Library planning is ADMIN_FACING but does not preserve this explicit no-routine-server-code administration requirement; implementation ownership still requires reconciliation with shared Control Center/System Administration surfaces.

RECOVERED · HI-FP-DATA-CONTRACT-CONSISTENCY-20260516-001

ForPrint canonical exchange forms and structured definitions should be tested for cross-module data sufficiency and consistency before publication, so data accepted at intake is also adequate for the downstream accounting, warehouse, production and related processes that consume it.

Context: Recovered from direct owner turn 1 requirement that every important step be covered by tests checking whether incoming fields are sufficient and mutually consistent across downstream operational uses. Current generic test governance is reused, while semantic ownership is reconciled with Contract Registry/Gateway/domain owners.

RECOVERED · HI-FP-CONTRACT-DUAL-FORM-20260516-001

External partner integrations should receive a clear human-readable form or specification together with an equivalent machine-readable contract or payload format, with explicit versioning so people and systems share one unambiguous exchange model.

Context: Recovered from the owner partner-integration example in turn 1: one human-understandable form plus one technical/machine form such as JSON. This is preserved as cross-module intent, not as proof that Library remains the current contract authority.

RECOVERED · HI-FP-GOV-IDEMPOTENT-GENERATED-20260529-001

Generated coordination snapshots, sync outputs and completion automation must be semantically idempotent: repeating the same verification or generation against unchanged semantic inputs must not create tracked Git differences solely because timestamps or equivalent non-semantic metadata were regenerated.

Context: Recovered from the Operational Registry historical source. The operator relayed a Blueprint clarification requiring idempotent sync snapshots and completion automation and explicitly called out timestamp-only churn. The source filename is only a 2026-05-29 label; visible history extends through 2026-06-19, so no exact logical date is asserted for this turn.

RECOVERED · HI-FP-LOGISTICS-REPORT-VISIBILITY-20260610-001

Human-facing module/check reports should provide an immediately scannable table in which each named check has an explicit PASS/WARN/FAIL result with clear green/yellow/red visual semantics; raw log lines remain evidence but should not be the only operator-facing summary.

Context: Recovered from the long-running Logistics dialogue. The operator explicitly requested a full visual table so successful, warning-level and failed checks can be distinguished immediately. This refines the existing unified-reporting direction without changing machine completion-envelope authority. The source filename is only a 2026-06-10 label and is not treated as the exact logical date of the turn.
Roadmap: BP-H10-UNIFIED-REPORTING

AGREED · HI-FP-BLUEPRINT-20260925-DETERMINISTIC-PROMOTION-001

ForPrint should maintain a permanent Deterministic Promotion Loop for every capability that uses an AI/LLM worker: observe real worker handoffs, identify recurring stable patterns, check for existing reusable deterministic capabilities,

and move economically/safely formalizable work into deterministic business logic, scripts, contracts, templates or bounded services after owner/contract/roadmap approval; the loop has no terminal DONE state and AI remains available for genuinely generative or ambiguous work.

Context: Owner confirmation on 2026-09-25 promotes and extends the earlier Managed Assistant Autonomy self-optimization loop into a cross-module architecture requirement.

Roadmap: deterministic_worker_promotion_policy_v0_1

AGREED · HI-FP-ARCH-20260926-001

ForPrint planning must keep three horizons distinct: a living practical roadmap for real near-term execution, a more detailed farther practical horizon, and a strategic horizon that preserves long-range theses without turning them directly into executable work.

Context: Owner clarified the planning model during the 2026-09-25/26 evening review; strategic material must be re-evaluated before future promotion into execution.

AGREED · HI-FP-ARCH-20260926-002

Internal assistant/developer service improvements such as startup-package quality and current-position projection should accumulate into a bounded improvement pool rather than fragment into immediate micro-tasks.

Context: These improve project operating infrastructure rather than ForPrint business capability; design must be re-evaluated when a material batch is eventually scheduled.

AGREED · HI-FP-ARCH-20260926-003

Reliable agent-mediated development should be reasoned about as Direction → Context → Authority → Execution → Verification → Acceptance → Evidence, with Acceptance kept distinct from technical verification.

Context: The evening review consolidated earlier worker/governance discussions into a long-range execution-control model.

AGREED · HI-FP-ARCH-20260926-004

ForPrint governance should distinguish the rights to propose, prepare, execute, verify, recommend acceptance, accept, release and escalate; performing work or producing evidence must not automatically grant acceptance authority.

Context: Accepted as a strategic governability principle; exact decision-right schema remains intentionally unfrozen.

AGREED · HI-FP-ARCH-20260926-005

ForPrint should eventually govern evolution through explicit generations or epochs so shared behavioral models can be piloted, stabilized, documented and rolled out dependency-aware across modules while legacy behavior is explicitly deprecated or retired.

Context: Artifact versions may advance at different rates; compatibility with the active generation matters more than equal version numbers.

AGREED · HI-FP-ARCH-20260926-006

ForPrint must preserve the security distinction that content is not authority, an instruction is not authorization, authorization is not physical capability, and untrusted or informational content cannot directly create privileged execution rights.

Context: Worker reasoning is useful but is not the sole trust boundary; system enforcement must remain effective even when a worker follows a bad instruction.

AGREED · HI-FP-ARCH-20260926-007

External-facing modules, especially Telegram Bot, Website, Calculator input surfaces and future public channels, require high-importance module-specific security against malicious inputs and recurring adversarial regression testing rather than a one-time audit.

Context: Threat families include prompt/instruction injection, malicious embedded content, SQL/command injection, path/schema/MIME abuse, resource exhaustion and later channel-specific attack classes.

AGREED · HI-FP-ARCH-20260926-008

Cross-module contract assurance should eventually include explicit consumer-declared expectations in addition to provider schemas; provider owns its interface, consumer owns its expectation, and syntactic compatibility alone does not prove semantic correctness.

Context: Accepted as a strategic direction for future Contract Registry / Integration Gateway maturity.

AGREED · HI-FP-ARCH-20260926-009

A generic green PASS must not be treated as universal proof of correctness; future verification evidence should be typed and preserve verifier identity/version, oracle provenance/validity, scope, feedback role and evaluation independence.

Context: Structural, contract, behavioral, governance, security, recovery and other evidence classes remain candidate taxonomy; Acceptance is a separate lifecycle decision.

AGREED · HI-FP-ARCH-20260926-010

ForPrint should evolve toward a Trustworthy Change lifecycle that separates intent, authorization, execution, verification, integration, acceptance and operational proof, with explicit residual-risk ownership and multi-anchor acceptance for suitable cross-domain changes.

Context: Exact state-machine details remain strategic design input and are not execution authority.

AGREED · HI-FP-ARCH-20260926-011

Assistants should not receive broad database access by default; domain owners should expose governed capabilities over canonical data, with invocation-time authorization based on identity, purpose, context and data scope where appropriate.

Context: The transport technology is intentionally not fixed; the domain-ownership and capability-mediated access principle is the strategic decision.

AGREED · HI-FP-ARCH-20260926-012

ForPrint project knowledge should prioritize machine-readable indexes and queryable dependency/flow graphs over exhaustive prose, while human documentation concentrates on intent, policy, rationale and ADR-like decision history.

Context: Use local module truth with central Blueprint projection and generate C4-like or task-specific views when useful; do not create a new documentation/graph module now.

AGREED · HI-FP-ARCH-20260926-013

CI should later become machine enforcement of stable ForPrint governance checks, while preserving CI PASS ≠ ACCEPTED ≠ MERGED ≠ RELEASED; CD belongs to a later maturity stage after release authority, staging, observability and rollback/recovery are mature.

Context: Do not prioritize CI/CD implementation before the initial worker/governance baseline and current roadmap rebuild are sufficiently stable.

AGREED · HI-FP-OPERATOR-CONSOLE-20260929-001

ForPrint should establish an Operator Console as a centralized Human Owner control surface for module work from phone or laptop, while keeping the durable execution, session, ownership, artifact, interrupt and recovery infrastructure in a separate Operator Control Plane / Execution Fabric rather than coupling it permanently to one UI.

Context: The Console is temporarily a co-equal high-priority planning direction with completion of module inventory because it can materially improve overall project throughput when the Human Owner is away from the local terminal.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-OPERATOR-CONSOLE-20260929-002

At most one active writable actor may own a module repository at a time. Internal Worker and Operator Assistant workspaces must not write concurrently to the same module; transfer uses controlled interrupt, durable checkpoint, seal/suspend, release and only then new acquisition.

Context: Operator Assistant may request Human Owner-authorized interruption of an Internal Worker. Internal Worker may request access to a Human Owner interactive session but must not preempt it automatically.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-SANDBOX-TOOLING-20260929-001

Isolated development sandboxes should encourage modern tools and dependencies when they materially improve quality, speed, observability or reliability; sandbox freedom should be broad while canonical repository, host, production secret, remote Git and external mutable-system boundaries remain more strongly controlled.

Context: This direction responds to observed development friction when assistants repeatedly worked around missing tooling instead of proposing useful tools such as ripgrep or browser/visual inspection support.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-SANDBOX-TOOLING-20260929-002

Material tools or dependencies added during sandbox work should be captured as Environment Delta evidence; repeatedly useful additions should be reviewed for promotion into canonical developer/module profiles, bootstrap capabilities or a discoverable Tool/Capability Registry rather than being repeatedly rediscovered.

Context: Fresh workers and assistants should know what tools and Makefile workflows already exist.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-EXECUTION-RECOVERY-20260929-001

Worker and operator-assistant execution must be designed for abrupt power loss through durable checkpoints and recoverable execution context so a restarted process can resume from a recent bounded state rather than reconstructing long work from zero.

Context: Unexpected server power loss is a real operating condition for the project and must be treated as a reliability requirement, not a theoretical edge case.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-ARTIFACT-PROVENANCE-20260929-001

Temporary generated artifact payloads may expire, but durable provenance should preserve enough metadata to identify the producing execution/module/base HEAD, source inputs, generator or recipe, file identity/hash and retention class so important outputs can be understood or recreated later.

Context: Binary payload retention and durable knowledge retention are separate concerns.

Roadmap: operator_console_control_plane_program_v0_1

AGREED · HI-FP-INVENTORY-LEGACY-CAPABILITY-20260929-001

Module inventory and later recurring audits must not classify old code only as used or unused: a legacy implementation may contain a valuable capability idea absent from the current roadmap. Such ideas should be raised for reconsideration and, when useful, preserved in roadmap/strategic planning even if the old code itself is later retired.

Context: This is a repeatable inventory requirement for L0/L1/L2, recovery audits, major refactors and later drift reassessment rather than a one-time Calculator-specific exception.

AGREED · HI-FP-EXECUTION-CONTROL-20260930-001

ForPrint should gradually acquire a persistent Execution Control Plane that knows declared work and reconciles it with durable evidence, instead of reconstructing operational state afterward from Git history, dirty files, old scripts or individual assistant memory.

Context: Git, tests, files, heartbeats, workflow results and artifacts are evidence; they are not by themselves sufficient semantic execution truth.

Roadmap: execution_control_plane_program_v0_1

AGREED · HI-FP-STABLE-WORKFLOW-20260930-001

Human Owner and future assistants should use stable supported workflow entrypoints instead of rediscovering script paths, argument combinations and invocation order in each session; Makefile, future CLI and Operator Console should call the same reusable backend workflow.

Context: When a recurring workflow fails because its reusable interface, ordering, projection, helper or recovery behavior is defective, repair that reusable workflow and add regression evidence where practical rather than fixing only the current session.

Roadmap: execution_control_plane_program_v0_1

AGREED · HI-FP-EXECUTION-ACTORS-20260930-001

Internal Worker, Operator Assistant and human-terminal execution should share one execution state model, module-lease semantics, checkpoint/status model and durable execution identity rather than becoming three separate operational realities.

Context: human_terminal explicitly covers the current bounded workflow where ChatGPT prepares a command or script, Human Owner executes it on the server and returns evidence.

Roadmap: execution_control_plane_program_v0_1

AGREED · HI-FP-EXECUTION-PHASING-20260930-001

Describe the full Execution Control Plane architecture now, but implement only a bounded minimum first; completing that minimum must be recorded as partial capability progress while all approved deferred architecture remains visible for later deliberate promotion.

Context: CF-10 owns the first contour; CF-12 carries intentionally deferred long-horizon capability. Deferred does not mean forgotten.

Roadmap: execution_control_plane_program_v0_1

AGREED · HI-FP-WORKER-SELF-HARDENING-20260930-001

After the first stable governed Worker attempts, progressively assign the Worker bounded, observable Blueprint self-hardening tasks so it improves the workflow/control environment used by later workers; start with lower-risk discovery/certification tasks and increase architectural complexity only from measured evidence.

Context: The self-hardening backlog is a governed source of candidate work, not automatic activation authority, and it must not expand the currently frozen Slice B2 implementation scope.

Roadmap: execution_control_plane_program_v0_1

AGREED · HI-FP-OPERATOR-CONSOLE-PHASING-20261001-001

Keep OC-01 as one compact roadmap step but deliver it through two explicit progressive contours, OC-01-MINI and OC-01-FULL. Both contours preserve governed Protected Terminal access and isolated Assistant Dev Sandbox work; detailed ordered implementation belongs in the implementation prompt rather than expanding the roadmap into a long checklist.

Context: MINI must already be genuinely useful from phone/laptop and prove one terminal cycle plus one sandbox-to-canonical cycle. FULL extends the proven vertical slice with richer profiles, resumable sandbox lifecycle, hardened promotion/recovery/audit and mature operator controls. Activation remains a separate Human Owner decision.

AGREED · HI-FP-EXTERNAL-AI-CAPABILITY-20261001-001

External customer-facing AI assistants may later become consumers of governed ForPrint domain capabilities for product discovery, calculation, technical requirements, logistics and related services; this refines existing HZN-014 rather than creating a new executable roadmap item.

Context: MCP is a current protocol example only. Preserve ESR-20260926-GAP-007: MCP, HTTP, RPC or another transport remains intentionally unfrozen. Revisit after the ecosystem has first stable working results and during deliberate strategic-goal review. Current implementation focus remains Operator Console.

AGREED · HI-FP-ARCH-EXECUTION-MODEL-20261007-001

Before broad module activation, ForPrint must integrate capability, ownership, roadmap decomposition, dependency, interface, workflow, verification and dispatcher models into one traceable Architecture & Execution Model portfolio so assistants and workers can discover existing and planned functionality, dependencies, supported workflows and completion criteria without reconstructing them from chat history.

Context: Human Owner approved this as the next cross-cutting architecture direction after the current OC-01 work. Existing Blueprint/CF10 primitives must be reused or extended first; rich control metadata should preserve a shallow execution path.
Roadmap: blueprint_architecture_execution_model_integration_v0_1

AGREED · HI-FP-POST-OC01-AEM-GATE-20261007-001

When OC-01-FULL reaches its canonical completion meaning, the next relevant portfolio architecture priority is AEM-01 of the Architecture & Execution Model integration initiative; activation remains an explicit Human Owner decision and must not reorder CF10/Control Foundation dependencies, preempt an active bounded writer, or convert the separate OC-02 future target.

Context: This durable priority pointer prevents the next architecture task from being lost during assistant replacement while keeping roadmap dependency and execution authority separate from prioritization.
Roadmap: blueprint_architecture_execution_model_integration_v0_1

Module — calculator_engine

Intent count: 33

AGREED · HI-CALCULATOR-ENGINE-001

Calculator Engine має бути головним власником canonical product/job calculation specification: продукт, параметри, матеріали, відходи, технологічні операції, ціна і розрахунковий час.

Context: Він не повинен перетворюватися на CRM, Warehouse чи Accounting.

AGREED · HI-CALCULATOR-ENGINE-002

Ручне замовлення з filename, hot-folder, CSV, операторської форми або legacy-джерела повинно нормалізуватися в той самий canonical request, що й онлайн-калькулятор.

Context: Не створюємо другу модель замовлення для ручного потоку.

AGREED · HI-CALCULATOR-ENGINE-003

Парсер filename/ручного вводу не має вгадувати неоднозначні параметри: при ambiguity він повинен вимагати clarification.

Context: Безпечніше запитати, ніж тихо підставити неправильний матеріал чи тираж.

AGREED · HI-CALCULATOR-ENGINE-004

Після стабілізації canonical schema потрібно зробити versioned filename grammar/parser і hot-folder workflow.

Context: Спочатку схема, потім адаптери.

AGREED · HI-CALCULATOR-ENGINE-005

Calculator формує canonical Job/Order Specification, який далі використовують Operations Control Registry, Prepress, Warehouse і production.

Context: Паперові/екранні представлення не повинні дублювати business truth.

AGREED · HI-CALCULATOR-ENGINE-006

Нормативні setup/waste/time rules повинні бути доступні Calculator для оцінки ціни, матеріалу та дедлайну.

Context: Фактичні production дані можуть пропонувати зміну норми, але не переписувати її автоматично.

RECOVERED · HI-CALCULATOR-ENGINE-007

Website і Mobile повинні вбудовувати Calculator/configurator без дублювання формул та цін у frontend.

Context: Це вже фіксувалося в попередньому портфоліо.

RECOVERED · HI-CALCULATOR-ENGINE-008

Calculator споживає canonical product/material/machine and print-mode definitions з Library через стабільний контракт.

Context: Попередні технічні матеріали явно відокремлювали Library reference input від pricing logic.

AGREED · HI-CALCULATOR-ENGINE-009

Для товарів на кшталт футболок, кепок та інших персоналізованих виробів потрібні зручні графічні інструменти/конструктори, які мають бути прив'язані до product configuration.

Context: Це людська вимога до майбутнього UX калькулятора, а не лише технічний configurator.

GAP · HI-CALCULATOR-ENGINE-010

Ми раніше називали конкретні зовнішні ресурси/сервіси як референси для конструкторів футболок/кепок і навіть визначали пріоритетний. У доступному корпусі файлів точні назви/URL зараз не відновилися.

Context: Це реальний documentation gap. Його не можна замінювати вигаданими назвами; перед реалізацією треба відновити точний список.

AGREED · HI-CALCULATOR-ENGINE-011

Референсні зовнішні сервіси потрібні не для сліпого копіювання, а як орієнтир по UX, поведінці configurator-а, прев'ю і роботі з зонами друку.

Context: У roadmap треба зберігати і самі URL/назви, і що саме з них беремо як орієнтир.

AGREED · HI-CALCULATOR-ENGINE-012

Кожен затверджений UX-reference повинен мати priority: primary reference / secondary reference / inspiration only.

Context: Щоб виконавець не змішував рівнозначно різні джерела.

AGREED · HI-CALCULATOR-ENGINE-013

Якщо в розмові названо конкретний зовнішній сайт/інструмент як референс, його назва й URL повинні потрапити в human-intent entry без узагальнення.

Context: Саме цього зараз не вистачило для calculator design references.

AGREED · HI-CALCULATOR-ENGINE-014

Calculator is the primary implementation owner for parsing, normalizing and rendering human production filenames, while Library owns the vocabulary/profile semantics.

Context: One generator/parser should prevent different modules inventing different naming rules.

Roadmap: CALC-FILE-01

AGREED · HI-CALCULATOR-ENGINE-015

Canonical human production filenames use single underscore semantic blocks; mandatory double-underscore separators are rejected as visually unnecessary.

Context: Human typing speed and readability are first-class requirements.

Roadmap: CALC-FILE-02

AGREED · HI-CALCULATOR-ENGINE-016

In the production filename quantity expression copies x N, N means physical sheets per copy, not PDF pages.

Context: The operator thinks in sheets; future validation must preserve this meaning.

Roadmap: CALC-FILE-03

AGREED · HI-CALCULATOR-ENGINE-017

Calculator should validate manually created filenames against deterministic file/job evidence and correct/report sheet-count mismatches when the production convention makes the result unambiguous.

Context: If the evidence is ambiguous, ask rather than silently guess.

Roadmap: CALC-FILE-04

AGREED · HI-CALCULATOR-ENGINE-018

For the Ricoh sheet profile, lamMat is the short human token for the agreed default matte 25 µm roll lamination without forcing the operator to type 25.

Context: Canonical database state still stores the exact lamination properties.

Roadmap: CALC-FILE-05

AGREED · HI-CALCULATOR-ENGINE-019

Manual/offline production files remain supported: after systems recover, Calculator parses and reconciles them into canonical job/order flow while preserving original filename and provenance.

Context: Do not make server availability a requirement for emergency manual production.

Roadmap: CALC-FILE-06

AGREED · HI-CALCULATOR-ENGINE-020

Calculator should emit required production capability/profile context in structured data so an abstract file is not misinterpreted when removed from its legacy machine directory.

Context: Human filename does not need to carry a physical printer serial number.

Roadmap: CALC-FILE-07

AGREED · HI-CALCULATOR-ENGINE-021

Calculator Job Specification is the agreed first Contract Registry lifecycle pilot toward Operations Control Registry.

Context: The exact contract remains to be detailed when Calculator target architecture is sufficiently mature.

Roadmap: CALC-CONTRACT-01

AGREED · HI-CALCULATOR-ENGINE-022

Exact Calculator references are recovered: sborka.ua is primary; FastPrint Pro and Wolf calc-polygraphy are secondary.

Context: Comparative reference only; not canonical truth or mandatory clone.

Roadmap: CALC-REF-01

AGREED · HI-CALCULATOR-ENGINE-023

Exact constructor references include Vizitka universal creator, FATLINE constructor and Maikoff constructor.

Context: Use them to study reusable personalized-product constructor patterns.

Roadmap: CALC-REF-02

AGREED · HI-CALCULATOR-ENGINE-024

Calculator needs an early visual product/configuration interface so operator usability feedback arrives while rules are still evolving.

Context: Visual UI is an early development instrument, not end-stage cosmetics.

Roadmap: CALC-UI-01

AGREED · HI-CALCULATOR-ENGINE-025

Calculator should support structured outsourcing of jobs to approved print partners when internal capacity/equipment/staff cannot safely fulfill them.

Context: Partner-specific execution belongs behind provider adapters.

Roadmap: CALC-OUTSOURCE-01

AGREED · HI-CALCULATOR-ENGINE-026

If a partner has no usable API, controlled browser automation may be used with evidence, retry bounds and human escalation.

Context: Credentials remain in centralized secrets infrastructure.

Roadmap: CALC-OUTSOURCE-02

AGREED · HI-CALCULATOR-ENGINE-027

Calculator requires a deep historical capability/ownership inventory before the next mature portfolio because early project stages placed cross-domain functionality inside Calculator.

Context: Candidate capabilities must be classified KEEP/MOVE/MERGE/REWRITE/DEPRECATE/REMOVE rather than deleted blindly.

Roadmap: CALC-INVENTORY-01

RECOVERED · HI-CALC-PRICING-20260922-001

Calculator pricing must be data-driven rather than hard-coded in Telegram, Website or application code: prices, tiers, options and modifiers should be editable as governed calculation policy, including segment-specific B2B/B2C calculation/presentation inputs without making Calculator the owner of customer identity.

Context: Recovered from the long-running Calculator history. Current refinement: CRM/commercial context supplies approved customer/segment references; channels own visual rendering; Calculator owns deterministic pricing effects.

RECOVERED · HI-CALC-PRICING-20260922-002

Calculator pricing administration should support filtered bulk policy changes over selected product/rule groups with previewable scope rather than requiring manual row-by-row edits for every product variant.

Context: Recovered from the owner's explicit examples of group-wide percentage adjustments with exclusions and distinct male/female/black-product handling.

RECOVERED · HI-CALC-APPLICABILITY-20260922-001

Calculator applicability rules should scale by consuming Library-owned canonical product/material references and attributes such as material class and grammage/ranges, then applying calculation eligibility/rule logic without creating a second canonical product/material catalog inside Calculator.

Context: Historical Calculator used reusable type/category and density-range relationships. Current architecture keeps semantic definitions and user-facing labels in Library while Calculator owns calculation applicability.

RECOVERED · HI-CALC-ADMIN-20260922-001

Calculator needs a serious operator/admin surface for Calculator-owned pricing and calculation policy—tiers, modifiers, applicability rules, calculation settings and previews—but that surface must remain bounded and must not become the canonical admin for Library catalogs, CRM, Accounting, orders or the whole ForPrint ecosystem.

Context: Recovered from the historical request for a scalable admin while explicitly reconciling the later Blueprint ownership split that supersedes the earlier whole-project-admin idea.

RECOVERED · HI-CALC-QUOTE-CONTRACT-20260922-001

Calculator quote intake must fail safely and deterministically on malformed or conflicting requests, preserve request correlation, and define domain-level idempotency so the same key plus equivalent payload can reuse a result while the same key plus a different payload becomes an explicit conflict rather than a second calculation.

Context: Historical quote/API work exposed both the intended idempotency semantics and a test-state isolation failure. Current runtime transport/retry idempotency must still be reconciled with Integration Gateway ownership.

RECOVERED · HI-CALC-QUOTE-CONTRACT-20260922-002

Calculator quote results and failures should use stable versioned machine contracts with structured line-item calculation evidence, assumptions/provenance, stable error codes and request correlation, while also supporting human-readable rendering without making any channel-specific presentation the source of calculation truth.

Context: Recovered from historical human/external quote reports, line-item outputs and schema-versioned error responses; current Contract Registry/Gateway/channel boundaries govern transport and presentation.

Module — forprint_operations_assistant

Intent count: 20

AGREED · HI-FP-OPERATIONS-ASSISTANT-001

Operations Assistant має бути максимально простим shop-floor interface: телефон/планшет, QR, великі кнопки, мінімум тексту, пока-йоке.

Context: Працівник не повинен вчитися працювати зі складною ERP.

AGREED · HI-FP-OPERATIONS-ASSISTANT-002

Працівник є сенсором: швидко повідомляє факт/аномалію, а система валідовує і маршрутизує його до canonical owner.

Context: Assistant не стає Accounting/Warehouse/production truth.

RECOVERED · HI-FP-OPERATIONS-ASSISTANT-003

Потрібні guided forms і contextual knowledge для receiving, stock check, production checkpoints, QC та видачі.

Context: Це вже було в старому портфолію.

AGREED · HI-FP-OPERATIONS-ASSISTANT-004

Для оцінки кількості аркушів або дрібних однотипних предметів має бути практичний режим: поміряти лінійкою висоту стопки і оцінити кількість по calibration profile.

Context: Саме цей 'лінійка + висота стопки' контекст треба зберігати як людську тезу, а не стискати до абстрактного quantity estimation.

AGREED · HI-FP-OPERATIONS-ASSISTANT-005

Calibration profile для матеріалу/товщини зберігається канонічно в Library; Assistant використовує його як довідкове правило.

Context: Для кожної оцінки потрібні tolerance/confidence і можливість ручної корекції.

AGREED · HI-FP-OPERATIONS-ASSISTANT-006

Якщо вимірювання виходить за нормальний діапазон, Assistant має запропонувати next safe action: переміряти, розбити стопку, запросити керівника, перейти на вагу/поштучний контроль.

Context: Не давати впевнене число при поганій якості вимірювання.

AGREED · HI-FP-OPERATIONS-ASSISTANT-007

Worker feedback по production variance має бути дуже коротким: normal / трохи вище / x2 / x5 / severe / machine problem або короткий voice note.

Context: Детальне виробниче обліковування не можна перекладати на виконавця.

AGREED · HI-FP-OPERATIONS-ASSISTANT-008

На поточному кроці працівник може запитати 'що мені робити?', а Assistant має показати інструкцію з урахуванням order/product/machine, material and role context.

Context: Перевага: текст -> картинки/схеми -> анімація -> реальне відео.

AGREED · HI-FP-OPERATIONS-ASSISTANT-009

Assistant повинен показувати STOP/hold події вище звичайної черги і вимагати acknowledgement, якщо працівник є адресатом критичної події.

Context: Скасоване замовлення не повинно тихо залишатися в списку робіт.

AGREED · HI-FP-OPERATIONS-ASSISTANT-010

Усі physical observations повинні мати evidence там, де це практично: фото, timestamp, device/operator, quantity/measurement context.

Context: Щоб пізніше можна було відновити, звідки взялася цифра.

AGREED · HI-FP-OPERATIONS-ASSISTANT-011

Лінійка/вага/камера/QR — це різні sensing methods; система повинна обирати найпростіший достатньо точний метод під конкретний матеріал.

Context: Не кожен інвентаризацію треба робити однаково.

AGREED · HI-FP-OPERATIONS-ASSISTANT-012

Operations Assistant should teach a new worker how to read the short production filename and which material/process each token implies in the active profile.

Context: Use text first, then schemes/reference images, later real equipment photos when useful.
Roadmap: OA-TICKET-01

AGREED · HI-FP-OPERATIONS-ASSISTANT-013

Operations Assistant displays the current Job Ticket on phone/tablet and can request another paper copy if the original is lost or damaged.

Context: OCR owns the ticket revision; Assistant is the human interface.
Roadmap: OA-TICKET-02

AGREED · HI-FP-OPERATIONS-ASSISTANT-014

Assistant production queue should expose structured priority and HOLD state with text/icon as well as color; color alone is insufficient.

Context: Urgent jobs must be obvious and held jobs must not look runnable.
Roadmap: OA-QUEUE-01

AGREED · HI-FP-OPERATIONS-ASSISTANT-015

Where Library has standardized finishing geometry, Assistant can show a visual final-result hint; otherwise an order-specific manager note/image can be displayed.

Context: Example: hole diameter and top-center position on a badge/tag.
Roadmap: OA-TICKET-03

AGREED · HI-FP-OPERATIONS-ASSISTANT-016

Assistant should expose a simple reprint reporting action from the Job Ticket/QR flow so the worker can report defect/reprint reason without learning a complex ERP.

Context: The resulting reprint semantics belong to OCR/Accounting/Warehouse.
Roadmap: OA-REPRINT-01

AGREED · HI-FP-OPERATIONS-ASSISTANT-017

Operations Assistant must provide equipment error/help lookup from parsed and indexed approved equipment documentation, including Ukrainian explanation and source illustrations when present.

Context: Exact error-code lookup should be preferred when a code exists.
Roadmap: OPS-EQUIP-01

AGREED · HI-FP-OPERATIONS-ASSISTANT-018

For production equipment, Operations Assistant must not invent repair procedures or freely search the internet for a repair answer; if approved documentation contains no answer it returns NOT_FOUND and escalates.

Context: Production-equipment troubleshooting is documentation-grounded only.
Roadmap: OPS-EQUIP-02

AGREED · HI-FP-OPERATIONS-ASSISTANT-019

Equipment-problem escalation is role-aware: a normal printer/operator can escalate to the production manager, while an authorized manager may create a service-support request through an approved channel.

Context: Permissions control who may contact the external service directly.
Roadmap: OPS-EQUIP-03

PROPOSED · HI-FP-OPERATIONS-ASSISTANT-020

Indexed manual procedures should progress from text and static figures to ordered step galleries and only later to optional animation where it adds real value.

Context: Animation is a later usability enhancement, not an early dependency.
Roadmap: OPS-EQUIP-04

Module — forprint_operations_control_registry

Intent count: 20

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-001

Operations Control Registry володіє operational order lifecycle після canonical specification від Calculator.

Context: Він координує виконання, а не рахує ціни й не веде бухгалтерію.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-002

Він резервує матеріал через Warehouse, веде execution gates і production queue.

Context: Гейти: READY_FOR_PRODUCTION, HOLD_PAYMENT, HOLD_MATERIAL, HOLD_APPROVAL, HOLD_CLARIFICATION, HOLD_PREPRESS.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-003

Job ticket містить order/revision/time/customer/product/qty/deadline/priority/material/operations/instructions/state/QR.

Context: Паперовий ticket — лише фізична projection.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-004

Скан старої ревізії job ticket повинен попередити про застарілий документ і запропонувати актуальну версію.

Context: Стара роздруківка не може дозволяти виконання старого плану.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-005

Критичні стани STOP_WORK_REQUIRED, CANCELLED_BY_CUSTOMER та EMERGENCY_HOLD йдуть окремим high-priority channel.

Context: Сповіщення мають мати audio + visual + acknowledgement + escalation.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-006

Кожна production action перевіряє current order state: перед START, перед auto machine command, на FINISH і перед handoff.

Context: QR ідентифікує замовлення, але ніколи не є permission token.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-007

Якщо cancellation приходить посеред операції, система фіксує фактично виконану частину, блокує handoff і записує physical WIP location/state.

Context: Не можна вдавати, що операція не починалась або повністю завершена.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-008

Для зупиненого WIP потрібен окремий event: order_id, operation_id, location_id, quantity_state, stopped_at, stopped_by, reason.

Context: Далі менеджер вирішує disposition: scrap/retain/rework/repurpose/give/transfer/other.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-009

Система може автоматично зупинити лише майбутні операції; доля вже виготовленого WIP — авторизоване управлінське рішення.

Context: Accounting окремо веде фінансові наслідки.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-010

Operations Control Registry тримає planned-vs-actual operational state по routing/operations, але physical material writeoff лишається у Warehouse.

Context: Чітко розвести production execution і складський truth.

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-011

Operations Control Registry owns the versioned Job Ticket as the complete human production document; filename remains a compact projection.

Context: Job Ticket can carry detailed operations, geometry, priority, holds, QR and revision.
Roadmap: OCR-TICKET-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-012

The future order/job identifier replaces the legacy dollar marker as the structured link between production file and canonical order/accounting flow.

Context: Dollar remains a legacy hint only.
Roadmap: OCR-FILE-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-013

Legacy wait should become explicit HOLD reasons such as HOLD_APPROVAL, HOLD_PAYMENT or HOLD_CLARIFICATION with START blocked until the gate clears.

Context: The file may be prepared in advance but cannot start printing while held.
Roadmap: OCR-HOLD-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-014

Legacy first should become structured expedite/priority class rather than hidden filename ordering.

Context: Several jobs may share the same high-priority class.
Roadmap: OCR-PRIORITY-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-015

Reprint must be a linked child-job workflow with reason, responsibility, quantity/material impact and separate billing consequence.

Context: Internal defect reprint may cost ForPrint without charging the customer.
Roadmap: OCR-REPRINT-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-016

Proof/sample purpose must be structured independently from billing policy; a prova/proof can be free, included or paid.

Context: Production purpose must not silently dictate accounting.
Roadmap: OCR-PROOF-01

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-017

A scanned stale Job Ticket/QR must warn that the paper revision is old and offer the current ticket; QR identifies the job but is not permission to execute.

Context: Current order state remains the execution gate.
Roadmap: OCR-TICKET-02

AGREED · HI-FP-OPERATIONS-CONTROL-REGISTRY-018

Stable shared Business Partner/person/organization and order/job identifiers should link Calculator, Accounting, Warehouse, Production, Logistics and CRM records.

Context: Operations Control Registry remains the current operational write/semantic boundary for shared party/order state.
Roadmap: OCR-DATA-01

RECOVERED · HI-OCR-ARCH-20260919-001

Historical asset retrieval needs explicit links to authoritative order, job and revision evidence so printed, approved, rejected or superseded status is not inferred from filenames or similarity alone.

Context: Recovered from the 2026-09-19 historical asset discussion and reconciled with the current OCR target-state direction.

AGREED · HI-OCR-ARCH-20260919-002

The first durable Process Manager capability is hosted inside Operations Control Registry as a distinct extraction-ready capability that owns process instances, current step, waiting conditions, expected events, timers, deadlines, retry/escalation state and transitions without absorbing foreign domain truth.

Context: The 2026-09-19 source required a hosted/extractable Process Manager but left the host open; later portfolio reconciliation selected Operations Control Registry as the working host.

Module — forprint_crm

Intent count: 24

AGREED · HI-FP-CRM-001

CRM має використовувати immutable person_id і organization_id; номер телефону — сильний практичний lookup, але не primary key БД.

Context: Людина може змінювати телефон, email чи месенджер.

AGREED · HI-FP-CRM-002

Одна людина може представляти різні організації в різний час; це temporal relationship з role, valid_from/to, status, source, confidence.

Context: При звільненні relation закривається, історія не стирається.

AGREED · HI-FP-CRM-003

Замовлення зберігає snapshot representation: person_id, organization_id, representation_id, confirmed_at.

Context: Щоб через рік було зрозуміло, від чийого імені робилося замовлення.

AGREED · HI-FP-CRM-004

Telegram або інший канал може один раз на замовлення уточнити, яку організацію людина зараз представляє.

Context: Це не повинно перетворюватися на нав'язливе підтвердження в кожному повідомленні.

AGREED · HI-FP-CRM-005

Потрібна generic anonymous one-off customer модель для випадків, де повний профіль не має сенсу.

Context: Не змушувати менеджера створювати фіктивні організації.

AGREED · HI-FP-CRM-006

CRM повинна вміти merge/split/dedup identity і зберігати cross-channel identifiers.

Context: Телефон, email, Telegram/Viber IDs — це identifiers, а не сама сутність людини.

RECOVERED · HI-FP-CRM-007

CRM — людський cockpit: open work, orders, blockers, communication context, next actions, dashboards і follow-ups.

Context: Вона не повинна ставати другою базою клієнтів/замовлень/платежів.

AGREED · HI-FP-CRM-008

CRM бере контактний контекст для AR collection, але сам фінансовий стан боргу належить Accounting.

Context: Менеджер бачить борг і діалог, але ledger truth не дублюється.

AGREED · HI-FP-CRM-009

One immutable person may simultaneously have a personal customer context and active representation of one or more organizations.

Context: A phone number identifies the person/contact path, not a single billing customer.

Roadmap: CRM-ID-01

AGREED · HI-FP-CRM-010

Organization representation remains temporal; when a person leaves an organization the relation closes for new orders while historical order snapshots remain intact.

Context: Do not let former employee orders fall onto the old organization.

Roadmap: CRM-ID-02

AGREED · HI-FP-CRM-011

An order must record the selected billing/customer context separately from the person who is communicating.

Context: The same person can order notebooks personally and company materials for Medicom using one phone number.

Roadmap: CRM-ID-03

AGREED · HI-FP-CRM-012

CRM should preserve normalized phone identity plus human display formatting and fast last-four lookup without using the phone as database primary key.

Context: Legacy folders/filenames may still show spaced phone formatting for operators.
Roadmap: CRM-ID-04

AGREED · HI-FP-CRM-013

CRM is a unified operational entry point: one guided interface may compose related actions across multiple owner modules.

Context: Humans should not manually visit many module tabs for one normal workflow.
Roadmap: CRM-WORKSPACE-01

AGREED · HI-FP-CRM-014

CRM should provide configurable role/department dashboards and large-screen wallboards for operational visibility.

Context: Keep dashboard capability encapsulated in CRM for now and extract later only if objectively justified.
Roadmap: CRM-DASH-01

AGREED · HI-FP-CRM-015

CRM may read/present cross-domain truth and submit structured commands, but it does not own Accounting, Warehouse, Production, Logistics or other foreign domain truth.

Context: Shared PostgreSQL does not remove logical ownership.
Roadmap: CRM-BOUNDARY-01

AGREED · HI-FP-CRM-016

Management/CRM views should surface plan-versus-actual variance for time, material, cost/price, reprints, delays and resource blockers.

Context: Configurable tolerance breaches become visible exceptions.
Roadmap: CRM-DASH-02

AGREED · HI-FP-CRM-017

A composite CRM workflow should support onboarding a new supplier/material from partial information and route enrichment/validation to domain owners.

Context: Enter known information once; gate only critical missing facts before irreversible actions.
Roadmap: CRM-WORKSPACE-02

AGREED · HI-FP-CRM-018

CRM should evolve toward Customer/Supplier 360, universal search, cross-module activity timeline, saved role workspaces and exception/task centers.

Context: These are views/projections, not a competing master database.
Roadmap: CRM-WORKSPACE-03

AGREED · HI-FP-U92-20260903-002

Customer Cabinet/Portal and basket/checkout composition belong in the CRM/business-workflow surface while Calculator, IAM, Accounting, Operations and Logistics retain their own semantic truth.

RECOVERED · HI-CRM-ARCH-20260919-001

Customer intelligence should separate a manager-selected Seed Profile, an evidence-derived Observed Customer Model and the currently applicable Effective Policy; a seed preset is not permanent classification or authority.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-CRM-ARCH-20260919-002

Customer behavior learning should distinguish baseline, exception, possible shift, probable shift and new baseline using recency, minimum evidence, hysteresis and contextual segmentation rather than lifetime-majority counting.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-CRM-ARCH-20260919-003

Descriptive customer evidence, customer preferences and ForPrint control policy are different layers; observed behavior must not silently grant commercial, financial or operational authority.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-CRM-ARCH-20260919-004

Customer health should be a multidimensional trajectory over owner-sourced evidence such as payment health, value, growth, margin, service burden, order clarity, dispute risk, relationship stability and strategic potential.

Context: Recovered from the 2026-09-19 evening architecture handoff; track movement, not only a static category.

RECOVERED · HI-CRM-ARCH-20260919-005

Customer-model and policy review should combine event-driven updates with periodic reconciliation and preserve transition history, evidence and reasons instead of overwriting the previous profile state.

Context: Recovered from the 2026-09-19 evening architecture handoff.

Module — forprint_accounting_registry_service

Intent count: 13

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-001

Accounting Registry Service є owner accounts receivable workflow: invoice/order, amount due/paid, due date, payment status, overdue days, promise-to-pay date.

Context: CRM і Telegram лише допомагають комунікувати.

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-002

Базовий state machine: DUE -> OVERDUE_SOFT -> OVERDUE_REMINDER -> PROMISE_TO_PAY -> WAITING_PROMISED_DATE -> OVERDUE_ESCALATED -> HUMAN_ATTENTION.

Context: Клієнтська відповідь змінює стан workflow, а не просто додається коментарем.

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-003

Side states: DISPUTED, PARTIAL_PAYMENT, PAYMENT_PENDING_RECONCILIATION, CONTACT_UNAVAILABLE, PAUSED, PAID.

Context: Щоб не штовхати кожен кейс в одну лінійну схему.

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-004

AI може адаптувати тон нагадування, але не може самостійно блокувати клієнта, зупиняти production/shipping, скасовувати кредит чи списувати борг.

Context: Це рішення людини/політики.

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-005

Cadence нагадувань має бути конфігурованим: quiet hours, max per day, importance/history і поточний контекст клієнта.

Context: Не перетворювати автоматизацію на спам.

RECOVERED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-006

Для procurement/goods receipt потрібен intake з Excel/CSV/PDF, OCR для фото/сканів і focused operator review.

Context: Попередній портфель уже містив цей напрям.

RECOVERED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-007

Supplier alias learning і fuzzy candidate mapping корисні, але OCR/matching не повинні ставати silent truth.

Context: Підтверджені mappings можна перевикористовувати.

AGREED · HI-FP-ACCOUNTING-REGISTRY-SERVICE-008

Фактична production variance впливає на actual cost analytics через Accounting, але первинні physical facts приходять від production/Warehouse.

Context: Не переносити production measurement у бухгалтерський модуль.

AGREED · HI-FP-ACCOUNTING-REGISTRY-009

Reprint responsibility and customer billing consequence are separate fields: an internal-defect reprint may consume material/cost without a new customer charge.

Context: Accounting records the financial consequence; OCR owns the reprint execution workflow.

Roadmap: ACC-REPRINT-01

AGREED · HI-FP-ACCOUNTING-REGISTRY-010

Proof/sample purpose does not automatically mean free; billing policy can be FREE, INCLUDED or PAID per case/policy.

Context: Do not infer finance solely from the legacy proba token.

Roadmap: ACC-PROOF-01

AGREED · HI-FP-ACCOUNTING-REGISTRY-011

Accounting should support incoming supplier-document parsing for Excel, PDF, Word and scans/photos with human confirmation when financial facts are uncertain.

Context: Repeated supplier templates should become increasingly deterministic.

Roadmap: ACC-GOODS-01

AGREED · HI-FP-ACCOUNTING-REGISTRY-012

Supplier-specific descriptions/part numbers map to one canonical material while preserving supplier alias/SKU provenance.

Context: The organization's preferred material identity remains stable.
Roadmap: ACC-GOODS-02

AGREED · HI-FP-ACCOUNTING-REGISTRY-013

A mature Accounting roadmap may include bounded preauthorized conditional payment mandates with strict limits, idempotency and audit.

Context: Early stages require confirmation; auto-payment is later high-risk automation.
Roadmap: ACC-PAYMENT-01

Module — forprint_semantic_retrieval_service

Intent count: 12

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-001

Ключовий принцип Semantic Retrieval: search finds candidates, domain owner decides truth.

Context: Пошук не може сам створити canonical клієнта, матеріал чи замовлення.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-002

Сервіс має бути cross-cutting: exact IDs + structured filters + lexical/BM25/trigram + text embeddings + image embeddings + domain/ACL filters + reranking.

Context: Гібридний пошук, а не лише vector DB.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-003

Search projection містить entity_id, entity_type, source_module, searchable_text, structured attrs, embedding/media refs, source_revision, indexed_at.

Context: Truth лишається у CRM/Warehouse/Library/OCR.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-004

Result state має бути явним: MATCHES / UNCERTAIN / NO_MATCH / POSSIBLE_NEW_ENTITY.

Context: Краще UNCERTAIN, ніж неправильний confident match.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-005

Warehouse use case: якщо QR зіпсований або відсутній, працівник описує предмет або фотографує його, а система показує кандидатів.

Context: Вибір остаточно підтверджує доменний owner/оператор.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-006

Telegram use case: клієнт надсилає фото і пише 'мені треба така штука'; Semantic Retrieval повертає candidates, Telegram уточнює і передає підтверджений контекст у Calculator.

Context: Це один із головних multimodal сценаріїв.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-007

Потрібен evaluation corpus: recall@5, precision, MRR, no-match accuracy, false-confident-match rate.

Context: Без вимірювання semantic search легко здається кращим, ніж є.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-008

ACL filtering має відбуватися до disclosure результатів.

Context: Навіть релевантний кандидат не можна показати користувачу без права доступу.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-009

Сервіс поки лишається PROPOSED_FOR_FORMAL_ADOPTION, не активний canonical module.

Context: Спочатку formal adoption у Blueprint окремим кроком.

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-010

Semantic Retrieval remains a noncanonical review-only candidate until its independent module value is proven.

Context: Do not promote it automatically because earlier artifacts exist.

Roadmap: SEM-REVIEW-01

AGREED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-011

Semantic retrieval may find fuzzy/semantic candidates such as a likely material or related document, but the relevant domain owner remains responsible for canonical truth and current state.

Context: Search finds candidates; domain owner decides truth.

Roadmap: SEM-BOUNDARY-01

PROPOSED · HI-FP-SEMANTIC-RETRIEVAL-SERVICE-012

Candidate value should be evaluated for multilingual/fuzzy material search and Project Inspector document/procedure similarity discovery before deciding PROMOTE, MERGE or RETIRE.

Context: Evaluation must compare the service with simpler alias/full-text/search approaches.
Roadmap: SEM-EVAL-01

Module — telegram_bot

Intent count: 45

RECOVERED · HI-TELEGRAM-BOT-001

Telegram має бути conversational operating interface: identity + memory + natural-language intent + multimodal + order/admin orchestration.

Context: Факти бере з canonical модулів.

AGREED · HI-TELEGRAM-BOT-002

Фото/опис товару йде в Semantic Retrieval, після чого Telegram показує кілька candidates і задає clarification.

Context: Не відповідати впевнено на основі одного fuzzy match.

AGREED · HI-TELEGRAM-BOT-003

Після підтвердження товару/варіанта Telegram передає structured request у Calculator.

Context: Chat history не є canonical order.

RECOVERED · HI-TELEGRAM-BOT-004

Speech-to-text повинен проходити через той самий intent pipeline, що й текст.

Context: Не створювати окрему бізнес-логіку для голосу.

RECOVERED · HI-TELEGRAM-BOT-005

Admin commands і money/destructive actions вимагають strong auth/confirmation.

Context: Наприклад taxi/procurement/status через owning modules.

RECOVERED · HI-TELEGRAM-BOT-006

Proactive events можуть повідомляти readiness/delay/courier/clarification, але лише з актуальних canonical states.

Context: Telegram не вигадує статус.

AGREED · HI-TELEGRAM-BOT-007

Для CRM identity Telegram може один раз у контексті замовлення підтвердити organization representation.

Context: Це допомагає зв'язати канал з CRM без ручного дублювання.

AGREED · HI-TELEGRAM-BOT-008

When a person can represent an organization, Telegram should lightly state the currently selected customer/billing context once per relevant order/dialogue.

Context: Example: 'оформляю замовлення на Medicom' so the person sees where the order will be billed.

Roadmap: TG-ID-01

AGREED · HI-TELEGRAM-BOT-009

Telegram must let the same person switch an order from organization context to personal context without creating a duplicate person.

Context: One channel identity may legitimately map to several business contexts.

Roadmap: TG-ID-02

AGREED · HI-TELEGRAM-BOT-010

Closed organization representations must not be proposed for new Telegram orders, while historical order snapshots remain reconstructible.

Context: Temporal CRM relationship drives channel suggestions.

Roadmap: TG-ID-03

AGREED · HI-TELEGRAM-BOT-011

Order-relevant Telegram confirmations should be persisted as structured/auditable evidence linked to the order, independent of whether the channel UI later still shows the message.

Context: A future retention/privacy policy must govern how much raw conversation is retained.

Roadmap: TG-AUDIT-01

AGREED · HI-TELEGRAM-BOT-012

Telegram should support bounded supplier/contact enrichment dialogues that collect missing structured information without repeated manual relaying.

Context: Confirmed facts are routed to the correct domain owner.
Roadmap: TG-SUPPLIER-01

AGREED · HI-TELEGRAM-BOT-013

Telegram roadmap should include voice-message transcription feeding the same structured intent pipeline as text, with future TTS/voice interaction later.

Context: Voice is a channel modality, not separate business logic.
Roadmap: TG-VOICE-01

AGREED · HI-TELEGRAM-BOT-014

Telegram Bot is the primary human communication orchestrator across customers, suppliers, carriers/taxi and other external contacts, while business decisions remain with the appropriate domain modules.

Context: Telegram communicates domain results rather than becoming the owner of Logistics/Accounting/etc.
Roadmap: TG-ROLE-01

AGREED · HI-TELEGRAM-BOT-015

Telegram must support both controlled menu-driven workflows for sensitive/financial interactions and a more natural conversational mode for ordinary communication.

Context: Sensitive flows use explicit identity/context/permissions and bounded choices.
Roadmap: TG-MODE-01

AGREED · HI-TELEGRAM-BOT-016

Before dispatching a task to Calculator, Logistics or another domain module, Telegram must collect the required input set for that module or explicitly route a structured missing-information state.

Context: Avoid wasteful request/clarification ping-pong caused by obviously incomplete tickets.
Roadmap: TG-TICKET-01

AGREED · HI-TELEGRAM-BOT-017

Completed and interrupted conversations must be preserved as traceable conversation episodes linked to participants and business context, while allowing multiple topics to coexist in one physical chat.

Context: Dialogue history may later be required for quality, disputes, delivery or accounting context.
Roadmap: TG-DIALOGUE-01

AGREED · HI-TELEGRAM-BOT-018

Telegram must actively close communication loops such as delivery handoff: notify, monitor expected arrival, warn about paid waiting/timeout when applicable, confirm receipt and escalate contact when necessary.

Context: Future voice/call adapter may be used for authorized escalation when text is not acknowledged.
Roadmap: TG-FOLLOWUP-01

AGREED · HI-TELEGRAM-BOT-019

Telegram requires a deep repository inventory and ownership reconciliation before the next mature portfolio because it is one of the oldest modules and may contain stale or cross-domain instructions/capabilities.

Context: Conflicting historical instructions must be classified and removed/reassigned only through reviewed migration.
Roadmap: TG-INVENTORY-01

AGREED · HI-TELEGRAM-BOT-020

External inbound messages must pass relevance/security/intake validation before they can trigger expensive search, AI dialogue or downstream module work.

Context: Accidental personal messages, pocket/noise text and suspicious extraction attempts must not become trusted tasks.
Roadmap: TG-INGRESS-01

RECOVERED · HI-TG-ARCH-20260919-001

Telegram may adapt tone, structure, length, formality and contextual explanation, but it must not silently alter domain facts such as price, quantity, deadline, status, payment requirement, delivery fact or guarantee.

Context: Recovered from the 2026-09-19 evening architecture handoff; communication presentation is separate from business truth.

RECOVERED · HI-TG-ARCH-20260919-002

Cross-module communication should prefer structured inbound domain requests/events and structured outbound communication intents; Telegram combines them with conversation state and communication context to render natural

language.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-TG-ARCH-20260919-003

Customer-specific terminology, recurring patterns, exceptions and communication preferences may inform Telegram behavior, but customer-specific semantics must not silently become global semantics.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-TG-ARCH-20260919-004

Ambiguous or multi-file conversations should converge toward a structured Order Draft through bounded clarification modes; Calculator, Library and other domain services remain the canonical constraint providers.

Context: Recovered from the 2026-09-19 evening architecture handoff; avoid endless free-text clarification loops.

RECOVERED · HI-TG-ARCH-20260919-005

Telegram should query a machine-readable historical asset index and deep-compare only a narrowed candidate set; it must not scan the raw customer archive live or own historical asset truth.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-TG-ARCH-20260919-006

Long-running process state, waiting conditions, timers, deadlines, retries and transitions must live outside Telegram; Telegram participates through communication intents, acknowledgements and clarification.

Context: Recovered from the 2026-09-19 evening architecture handoff.

RECOVERED · HI-TG-DATA-INGEST-20260922-001

Telegram dialogue history should be recoverable from exported customer-chat HTML through a deterministic parser that produces structured semantic input/response examples, merges short same-context messages where appropriate, and does not turn non-text media placeholders into dialogue examples.

Context: Recovered from the filename-label 07.10 Telegram data-ingest dialogue. The historical 20-column Supabase shape is evidence, not automatically the current schema.

RECOVERED · HI-TG-DATA-INGEST-20260922-002

Dialogue/corpus ingestion should resolve an existing customer identity reference rather than invent a new client identity from the chat export; current canonical customer identity remains outside Telegram Bot.

Context: Historical source explicitly maps client_id from an existing clients table. Current architecture refines the owner to CRM/identity surfaces.

RECOVERED · HI-TG-DATA-INGEST-20260922-003

Before expanding Telegram training data synthetically, use known-good examples that already import successfully to establish the active schema/format contract instead of guessing headers or field shapes.

Context: Owner explicitly changed the historical workflow to compatibility-first generation after providing known-good CSV samples.

RECOVERED · HI-TG-DATA-INGEST-20260922-004

Synthetic order_modification coverage should include realistic changes to an existing order such as add/replace/change, references to a previous order, and quantity or detail modifications rather than only new-order requests.

Context: Recovered directly from the historical synthetic-data expansion request.

RECOVERED · HI-TG-BEHAVIOR-20260922-001

Telegram intent/classification semantics should explicitly represent sales opportunity, customer refusal and objection-handling states so downstream dialogue logic can choose appropriate selling, clarification or fallback behavior instead of treating all requests as one generic dialogue class.

Context: Recovered from the internally anchored 2025-10-12 behavior-model dialogue. Exact current taxonomy names remain subject to live classifier/schema reconciliation.

RECOVERED · HI-TG-ML-20260922-001

Telegram classifier training data should remain modular by behavior/domain rather than be forced into one giant table, so weak domains such as sales, objections or order handling can be improved independently while the training pipeline combines them under a versioned dataset contract.

Context: Recovered directly from the owner discussion of separate dialogue/sales/objection/order datasets. Dataset registry, weighting and merge mechanics remain to be reconciled.

RECOVERED · HI-TG-ML-20260922-002

Recurring Telegram classifier evaluation must not rely indefinitely on one static test dataset that can leak into tuning; long-running quality evidence should include fresh unseen cases, provenance/leakage controls and an independent expected-label oracle or review path before automated retraining or promotion decisions.

Context: Recovered from the owner's explicit concern that DeBERTa can adapt to a repeatedly used test set. External LLM generation is only one historical candidate for fresh cases, not the oracle and not a required provider.

RECOVERED · HI-TG-HUMAN-TALK-20260922-001

Telegram human-talk response templates should use normalized intent/style/sentiment references and produce short, natural, non-verbatim-repetitive responses with multilingual support for Ukrainian, Russian and English; template priority remains a bounded ranking input rather than business truth.

Context: Recovered from the filename-label 12.10 human-talk/template dialogue. The owner explicitly moved the schema from textual style/sentiment fields to style_id/sentiment_id foreign-key references, required 1-2 sentence natural responses, uk/ru/en support and priority 0-10.

RECOVERED · HI-TG-HUMAN-TALK-20260922-002

Telegram template and synthetic-data generation must stay inside the current canonical intent, style and sentiment lookup vocabularies; generated rows must not silently invent new labels or expand the taxonomy without an explicit schema/taxonomy decision.

Context: The owner supplied explicit lookup tables. Later assistant-generated rows introduced unsupported friendly/playful/aggressive/urgent values; this is preserved as a historical contract-drift defect, not an authorized taxonomy expansion.

RECOVERED · HI-TG-ML-ENGINEERING-20260922-001

Before retraining Telegram classifiers, datasets should be audited for invalid labels, conflicting duplicate texts, ordinary duplicates, suspicious/low-quality examples, class imbalance and train/validation/test leakage; the resulting split and augmentation lineage should remain explicit.

Context: Recovered from the October-2025 Telegram ML engineering dialogue after owner-reported style accuracy 0.6394. The owner asked for dataset correction, balancing and detailed review before retraining.

RECOVERED · HI-TG-ML-ENGINEERING-20260922-002

Telegram intent, style and sentiment classifiers should follow one shared structural discipline across training, classification and evaluation while preserving entity-specific datasets, label contracts, model artifacts, revisions, reports and paths so one classifier domain cannot collide with another.

Context: Recovered from repeated owner requests to align intent/style/sentiment training and evaluation scripts without collapsing their separate artifact namespaces.

RECOVERED · HI-TG-ML-ENGINEERING-20260922-003

Existing Telegram classifier label-map files are durable compatibility contracts: training, classify and evaluate code should read rather than silently overwrite them, deliberately support documented name-to-id and id-to-name forms where required, and preserve downstream decoder compatibility.

Context: Recovered from the owner's explicit label-map preservation rule and historical intent-decoder failures.

RECOVERED · HI-TG-ML-ENGINEERING-20260922-004

Telegram ML paths and operational settings should be centralized and composed from canonical configuration roots rather than scattered hard-coded strings; config refactors must preserve existing consumers and classifier-domain-specific path constants must prevent intent/style/sentiment artifact collisions.

Context: Recovered from repeated config.py/.env/path refactoring requests and the historical generic DEBERTA_ARCHIVE_BASE_DIR collision.

RECOVERED · HI-TG-ML-ENGINEERING-20260922-005

Telegram classifier evaluation should emit both machine-readable metrics/artifacts and human-readable diagnostics, including per-class/confusion evidence where useful, stable artifact identities and explicit paths to created outputs so the operator can inspect results without losing reproducibility.

Context: Recovered from the owner's requests for readable Excel-oriented evaluation alongside JSON/confusion artifacts and predictable metrics/log filenames.

RECOVERED · HI-TG-BEHAVIOR-RUNTIME-20260922-001

Telegram behavior should be modeled and reviewed explicitly before production execution logic is allowed to drive architecture; synthetic phrases and scenario fixtures validate behavior specifications but must not silently become the source of production business rules.

Context: Recovered from the long-running Telegram history where the owner explicitly reset priority from classifier/runtime tuning back to behavior-model design and treated synthetic phrases as behavior fixtures.

RECOVERED · HI-TG-QUALIFICATION-20260922-001

Telegram may conversationally collect missing customer qualification data such as contact details, activity/product interests and preferred delivery method, but any durable profile update must be expressed as a confirmed write intent to the canonical CRM/identity owner rather than maintained as Telegram-owned customer truth.

Context: Recovered from historical client-qualification behavior and reconciled with current CRM identity ownership. The exact writable field contract remains to be defined with CRM.

RECOVERED · HI-TG-BEHAVIOR-REGISTRY-20260922-001

Telegram behavior specifications should have one structured canonical registry and may generate operator-friendly Excel/HTML views with integrity, duplicate and KPI/coverage indicators; generated views must remain derived projections rather than manually maintained competing sources of truth.

Context: Recovered from the historical behavior-registry workflow where Excel/HTML were requested as default operator artifacts and integrity/KPI/duplicate metadata were part of the registry.

RECOVERED · HI-TG-BEHAVIOR-RUNTIME-20260922-002

Telegram behavior registries, XLS workbooks, wizard simulators and other design/test artifacts must be clearly classified and separated from production runtime dependencies; scenario simulation should remain modular behind a small dispatcher while production runtime consumes only approved runtime contracts.

Context: Recovered from the historical ORDER/VENDORS wizard evolution and the owner's explicit separation of XLS/wizard development artifacts from the real Supabase-backed Telegram runtime.

RECOVERED · HI-TG-TOOLING-20260922-001

Telegram tooling that mutates generated workbooks, registries or other artifacts must verify the artifact-level postcondition after a claimed success; command exit success alone is insufficient when the expected file content or structural change did not actually occur.

Context: Recovered from a historical false-success candidate where a patcher reported success but the owner observed that the workbook had not changed.

Module — website

Intent count: 8

RECOVERED · HI-WEBSITE-001

Website є customer web channel і повинен використовувати ті самі canonical backend capabilities, не створювати власні ціни/матеріали/order truth.

Context: Це було в попередньому портфоліо.

RECOVERED · HI-WEBSITE-002

Website має вбудовувати Calculator configurator/quote flow без дублювання formulas.

Context: Frontend лише представляє конфігурацію і результати.

RECOVERED · HI-WEBSITE-003

Потрібні catalog, file upload/design editor integration/preview, checkout, account/status/repeat order.

Context: Старий портфель вже містив цей flow.

AGREED · HI-WEBSITE-004

Visual product designer для персоналізованих виробів має спиратися на затверджені референсні сервіси й canonical design-system правила.

Context: Точний список calculator design references зараз позначений GAP до відновлення.

RECOVERED · HI-WEBSITE-005

Website повинен використовувати canonical ForPrint Design System tokens/components/themes/density/accessibility.

Context: Є окремий Blueprint addendum про owner design system.

AGREED · HI-WEBSITE-006

Website не повинен зберігати business truth у frontend state; assets/order refs мають вести до owning modules.

Context: Особливо при refresh/retry.

RECOVERED · HI-WEBSITE-007

Потрібні explicit review/confirmation перед final submission і mobile/performance/accessibility considerations.

Context: Не жертвувати зрозумілістю заради складного конструктора.

AGREED · HI-FP-U92-20260903-001

Website is the public Web Presence / SEO Entry Point: public façade, discoverability and routing only; it must not become authentication, pricing, payment, basket or canonical order truth.

Module — mobile_app

Intent count: 6

RECOVERED · HI-MOBILE-APP-001

Спочатку треба підтвердити реальну цінність native app проти PWA/responsive web.

Context: Не будувати native лише тому, що 'має бути додаток'.

RECOVERED · HI-MOBILE-APP-002

Унікальна mobile value: push, camera, file share, device integration, low-friction operational/customer flows.

Context: Business logic не дублюється на device.

RECOVERED · HI-MOBILE-APP-003

MVP може включати catalog/configuration/calculation/order creation/status через shared APIs.

Context: Calculator/identity/API readiness — dependency.

AGREED · HI-MOBILE-APP-004

Для працівників shop-floor частина mobile UX може бути Operations Assistant surface, а не окремий паралельний бізнес-додаток.

Context: Треба не дублювати функції між customer app та internal assistant.

AGREED · HI-MOBILE-APP-005

Єдиний design language з Website/Library UI system є обов'язковим.

Context: Щоб не з'явилися три різні ForPrint UI.

AGREED · HI-FP-U92-20260903-003

Native Mobile App remains proposed/paused until customer scale, mobile traffic and native-only value justify implementation; responsive web and Customer Portal are the present priority.

Module — forprint_library

Intent count: 20

RECOVERED · HI-FP-LIBRARY-001

Library володіє stable canonical IDs і versioned schemas для products/materials/services/operations.

Context: Price/accounting/operational state від цього відділення.

RECOVERED · HI-FP-LIBRARY-002

Aliases, supplier part numbers, alternate names, normalization rules і provenance/confidence мають бути частиною semantic layer.

Context: Це дозволяє безпечно мапити зовнішні джерела.

RECOVERED · HI-FP-LIBRARY-003

External catalog ingestion повинен зберігати source/provider/fetched-at/key/hash/raw snapshot refs і давати human review для ambiguous merge.

Context: Не переписувати canonical truth без підтвердження.

AGREED · HI-FP-LIBRARY-004

Library є canonical owner calibration profiles для фізичних вимірювань Operations Assistant, включно з параметрами товщини/ваги/толерансів.

Context: Сам Assistant лише використовує профіль.

RECOVERED · HI-FP-LIBRARY-005

Library -> Calculator reference input має бути deterministic, typed, versioned і read-only.

Context: Pricing formulas залишаються в Calculator.

AGREED · HI-FP-LIBRARY-006

Library повинна зберігати SOP/instruction/media knowledge для contextual work guidance.

Context: Operations Assistant показує потрібну інструкцію з контексту.

RECOVERED · HI-FP-LIBRARY-007

Canonical ForPrint Design System owner закріплений за Library; Website та інші UI мають мігрувати до нього без page-local систем.

Context: Це вже було окремо додано в Blueprint.

AGREED · HI-FP-LIBRARY-008

Reference photos/product media можуть індексуватися Semantic Retrieval, але Library лишається owner canonical assets/metadata.

Context: Пошуковий індекс — projection.

AGREED · HI-FP-LIBRARY-009

Library owns canonical aliases, common misspellings, short production tokens, naming profiles and profile-specific defaults for materials/operations/device capabilities.

Context: Calculator consumes these definitions instead of maintaining a private vocabulary.

Roadmap: LIB-FILE-01

AGREED · HI-FP-LIBRARY-010

A short human filename may omit common information only when the active naming profile defines the default unambiguously.

Context: Example: Ricoh lamMat represents the agreed default thin matte roll lamination.

Roadmap: LIB-FILE-02

AGREED · HI-FP-LIBRARY-011

The same short token may have different meaning across equipment/production profiles, so structured profile/capability context must remain explicit.

Context: Do not globally interpret a token without its naming/production profile.

Roadmap: LIB-FILE-03

AGREED · HI-FP-LIBRARY-012

Library semantic changes that affect inter-module payloads must flow through a versioned Contract Registry/adoption process rather than being changed locally in one consumer.

Context: Avoid modules drifting onto different meanings of the same field/token.
Roadmap: LIB-FILE-04

AGREED · HI-FP-LIBRARY-013

Library should eventually provide standardized visual/SOP instructions for Job Ticket and Operations Assistant steps, with manager-provided notes/images as fallback when no standard exists.

Context: Human guidance is a Library knowledge responsibility; execution truth remains OCR.
Roadmap: LIB-FILE-05

AGREED · HI-FP-LIBRARY-014

Library is publication owner for shared ForPrint UI tokens, themes, reusable components, component catalog, versions and adoption metadata.

Context: Blueprint governs; consumers compose; Inspector validates.
Roadmap: LIB-UI-01

AGREED · HI-FP-LIBRARY-015

Shared indexes should help modules discover reusable technical primitives and semantic definitions before creating competing implementations.

Context: Domain business rules still belong to domain owners.
Roadmap: LIB-INDEX-01

AGREED · HI-LIBRARY-U131-20260906-001

ForPrint Library is the shared semantic and contract reference authority; modules should consult its existing canonical/recommended semantics before creating significant competing shared implementations.

Context: This extends the Library role already present in module policy; it does not move domain business ownership into Library.
Roadmap: LIB-SEMANTIC-REGISTRY

AGREED · HI-LIBRARY-U131-20260906-002

A first registered semantic ID may remain stable while maturity, recommended implementation names, aliases and contract versions evolve.

Context: Registration does not automatically mean CANONICAL.
Roadmap: LIB-SEMANTIC-REGISTRY

AGREED · HI-LIBRARY-U131-20260906-003

Semantic registration should prioritize module-public, cross-module, external-boundary, high-criticality and widely reused surfaces rather than every local helper.

Context: Interaction scope and criticality are independent signals for standardization review.
Roadmap: LIB-SEMANTIC-REGISTRY

AGREED · HI-LIBRARY-U131-20260906-004

Library should batch unresolved semantic registration/enrichment proposals and publish versioned semantic changes; Blueprint remains responsible for rollout timing and adoption mode across affected modules.

Context: Adoption may be informational, new-code-only, migrate-on-touch, roadmap-required or migration-required.
Roadmap: LIB-SEMANTIC-REGISTRY

RECOVERED · HI-LIBRARY-ARCH-20260919-001

Historical/reference asset indexing should use stable asset and revision semantics with provenance; an index/search result is a projection and must not become canonical asset, order or production truth.

Context: Recovered from the 2026-09-19 historical asset indexing discussion.

Module — forprint_prepress_hub

Intent count: 25

RECOVERED · HI-FP-PREPRESS-HUB-001

Prepress Hub має визначити file lifecycle від upload до production-ready, readiness statuses, blockers, warnings, fixable issues і evidence.

Context: Master file і derived previews потрібно розділяти.

RECOVERED · HI-FP-PREPRESS-HUB-002

Core preflight: format/pages/dimensions/orientation/resolution/bleed/color-space/fonts де це можливо.

Context: Порівнювати з Calculator job spec і Library requirements.

RECOVERED · HI-FP-PREPRESS-HUB-003

Safe previews/thumbnails і bounded normalization/conversion можуть бути автоматичними, але master не модифікується тихо.

Context: Потрібен before/after evidence.

RECOVERED · HI-FP-PREPRESS-HUB-004

Operator-assisted fix workflow потрібен для неоднозначних випадків.

Context: Автоматизація не замінює judgment там, де немає детермінованої відповіді.

AGREED · HI-FP-PREPRESS-HUB-005

Prepress blockers стають HOLD_PREPRESS у Operations Control Registry і блокують production start.

Context: Гейт має бути видимий в job ticket.

AGREED · HI-FP-PREPRESS-HUB-006

Production-ready package передає verdict + asset refs + revision, щоб стара версія файлу не пішла у виробництво.

Context: Revision consistency критична.

AGREED · HI-FP-PREPRESS-HUB-007

Prepress should expose a reusable deterministic document probe for PDF page count, dimensions and other technical evidence used by Calculator validation.

Context: Calculator initiates business validation; Prepress supplies file facts.

Roadmap: PP-FILE-01

AGREED · HI-FP-PREPRESS-HUB-008

Production filename sheet-count correction may be automatic only when the agreed production/PDF convention makes physical sheet count deterministic; otherwise clarification is required.

Context: Do not guess duplex/imposition meaning from ambiguous files.

Roadmap: PP-FILE-02

AGREED · HI-FP-PREPRESS-HUB-009

PDF production metadata may later carry stable production properties as an interoperability projection, but it does not replace canonical order/job state.

Context: JDF/XJDF/PDF metadata remain future adapter directions.

Roadmap: PP-INTEROP-01

AGREED · HI-FP-PREPRESS-HUB-010

Prepress Hub roadmap deepening must be recovery-first: recover all prior detailed operator discussions, project evidence and existing module capabilities before filling only genuine gaps with PROPOSED synthesis.

Context: The operator previously described this module in substantial detail and does not want that intent replaced by a fresh generic design.

Roadmap: PREPRESS-RECOVERY-01

RECOVERED · HI-PREPRESS-ARCH-20260919-001

Prepress should be able to emit machine-readable technical asset facts and fingerprints for historical indexing, including provenance and representative-page evidence, while not becoming the archive, customer-history or search-result authority.

Context: Recovered from the 2026-09-19 historical asset indexing discussion.

RECOVERED · HI-PREPRESS-CAPABILITY-20260922-001

Prepress development should be capability-first: first determine which practical file-processing tasks can actually be automated, which native tools already exist and which utilities ForPrint must build; only after that should broader cross-module integration architecture be designed.

Context: Recovered directly from the 27.06 owner discussion. This refines the broad May zero-platform proposal without discarding the canonical Prepress module itself.

RECOVERED · HI-PREPRESS-ASSISTANTS-20260922-001

Prepress assistants should remain product-specific where domain behavior differs—such as Photoshop, Illustrator, CorelDRAW and PDF/Acrobat—while reusable low-level file inspection belongs in shared fact-oriented tools that return stable structured technical evidence rather than making production decisions.

Context: Recovered from the owner's rejection of one abstract Raster Assistant and the accepted shared-tools model. Specialized assistants should reuse common inspection capabilities instead of duplicating them.

RECOVERED · HI-PREPRESS-CANDIDATES-20260922-001

A single input may legitimately produce several controlled Prepress processing candidates because different tools or strategies can yield different visual quality; candidate ordering must depend on file type, task and risk rather than a universal tool hierarchy, and each candidate should expose preview plus risk/confidence evidence.

Context: Recovered from the owner's bleed/edge example and explicit requirement to show different processing variants when multiple tools can produce materially different results.

RECOVERED · HI-PREPRESS-EXECUTION-20260922-001

Prepress execution should be Unix-first whenever deterministic inspection or preparation is practical, while Windows/Photoshop is a bounded quality executor for Photoshop-specific, layered or visually sensitive work; the workstation must not become the central orchestration brain.

Context: Recovered directly from the owner's request to keep practical work on Unix where possible and use the remote Windows graphics station only when its specialized tooling adds real value.

RECOVERED · HI-PREPRESS-RASTER-POLICY-20260922-001

Raster preflight should treat effective resolution and aspect suitability as separate checks and use configurable policy thresholds for minimum effective DPI and maximum automatic nonproportional adjustment; outside the allowed policy envelope the workflow must request confirmation or operator review instead of guessing.

Context: Historical examples mentioned about 25 DPI and 3-5% distortion, but those values are examples/configuration ideas rather than permanent production constants.

RECOVERED · HI-PREPRESS-RASTER-POLICY-20260922-002

Automatic K-only enhancement of thin dark raster text must be treated as a heuristic candidate with explicit confidence/risk gating and preview/manual fallback; automation must not blindly recolor dark image or design content.

Context: Recovered from the owner's real print-quality problem and the accepted safety boundary that rasterized text cannot always be distinguished reliably from other dark artwork.

RECOVERED · HI-PREPRESS-PDF-AUTOMATION-20260922-001

ForPrint should automate client file intake as far as safely possible: accept uploaded/SFTP-style files, validate them against product/prepress rules, apply deterministic low-risk corrections automatically, generate a visual proof for confirmation, ask structured clarification when required, and route cases that cannot be safely automated to manual paid prepress rather than silently guessing.

Context: Recovered directly from the PDF Assistant primary conversation. This extends current Prepress lifecycle, safe-preview and operator-fallback intent with an explicit online intake/proof/clarification loop.

RECOVERED · HI-PREPRESS-PDF-ENGINEERING-20260922-001

ForPrint-owned PDF automation should be Python-first for orchestration, rules, adapters and reporting; non-Python/native components such as C++ may be introduced only for bounded functions where a concrete performance or capability gap justifies the added complexity, rather than moving the whole PDF subsystem to another language.

Context: Recovered from the owner's explicit confirmation of the preferred engineering policy in the PDF Assistant conversation.

RECOVERED · HI-PREPRESS-PDF-STRATEGY-20260922-001

ForPrint should evaluate and plan a ForPrint-owned PDF Control Layer capable of gradually covering the business-critical callas-like automation needed by the printshop, while keeping external commercial or native PDF engines replaceable rather than making one vendor the owner of ForPrint PDF business logic; the exact scope must be reconciled and approved through Blueprint before implementation.

Context: Recovered from the owner's request to prepare a machine-oriented Blueprint prompt for an internal callas-like PDF project with roughly a year available for implementation. The source does not authorize a full clone of callas or a specific commercial-engine purchase.

RECOVERED · HI-PREPRESS-RASTER-PIPELINE-20260922-001

Raster client inputs such as PNG/JPEG and similar formats should be prepared for print through a bounded raster-processing stage, normalized to an appropriate PDF representation, passed through PDF-specific validation, and then handed to Prepress Hub for downstream readiness/production handling; this pipeline should preserve the separation between raster adaptation, PDF validation and overall Prepress orchestration.

Context: Recovered directly from the Prepress Assistant architecture primary source. Product-specific assistants and Unix/Windows execution separation are corroborating evidence already captured from the 27.06 source.

RECOVERED · HI-PREPRESS-OPERATOR-SURFACE-20260922-001

Prepress should provide one lightweight operator surface where a file can be drag-dropped or selected, a reusable ordered preset/chain can orchestrate the needed Acrobat/Photoshop/Illustrator or other product-specific processing steps, and the output destination can be defaulted or chosen; this surface is orchestration UX and must not collapse product-specific assistants into one universal domain brain.

Context: Recovered directly from CRM-source turn 1, which predates the later Prepress sources. Reconciled with the 27.06 owner decision that product-specific assistants remain separate while shared orchestration/tools may compose their work.

AGREED · HI-FP-PREPRESS-HUB-20260925-VISUALIZATION-DISCOVERY-001

Prepress Hub should treat automated product visualization as a future discovery-heavy capability: compare deterministic 2D mockups, scripted/3D rendering, AI-assisted visualization and hybrid pipelines, prefer deterministic rendering where adequate, and require a fresh technology review at the actual implementation date rather than freezing the 2026 tool stack.

Context: Owner confirmation on 2026-09-25: implementation may occur much later and must re-evaluate then-current tooling, fidelity, cost, licensing, privacy, local-vs-hosted execution and print-specific limitations.

Roadmap: PREPRESS_VISUALIZATION_DISCOVERY_20260925

AGREED · HI-PREPRESS-GDL-EMPIRICAL-LEARNING-20261001-001

Graphic Design Lab should learn from real customer requests, Creator prompts, Creator outputs, revision cycles, operator assessment and customer acceptance before broad design automation expands. Failed prompts and failed results remain learning evidence; one case may create a hypothesis but must not automatically create a global rule.

Context: The next bounded GDL contour adds a lightweight empirical-learning layer over the already verified Product Playbook, Guided Intake and Creator Handoff foundation. Repository data should be sanitized metadata and references rather than heavy design assets or customer PII.

Roadmap: prepress_gdl_creator_empirical_learning_foundation_v0_1

AGREED · HI-PREPRESS-GDL-CONTOUR-GOVERNANCE-20261001-001

Future GDL development should be released one reconciled execution contour at a time. Later contours preserve intent, rationale and dependencies but are not frozen implementation instructions; before every release Blueprint must reconcile current Prepress state, empirical findings, Human Intent and pending amendments.

Context: Sequence 1 is Creator Empirical Learning Foundation. Creator Result Package remains sequence 2. Evaluation taxonomy and prompt-pattern composition require empirical evidence. No later contour self-activates.

Roadmap: prepress_gdl_empirical_learning_horizon_20261001_v0_1

Module — warehouse_service

Intent count: 11

AGREED · HI-WAREHOUSE-SERVICE-001

Warehouse володіє physical stock truth, reservations і actual material writeoff.

Context: Calculator рахує потребу, але не списує склад.

AGREED · HI-WAREHOUSE-SERVICE-002

Operations Control Registry резервує матеріал через Warehouse і отримує HOLD_MATERIAL, якщо ресурс не готовий.

Context: Ніяких прямих прихованих змін stock з інших модулів.

AGREED · HI-WAREHOUSE-SERVICE-003

Semantic Retrieval допомагає знайти inventory item за описом/фото при відсутньому QR, але Warehouse/оператор підтверджує істину.

Context: Search лише кандидат.

AGREED · HI-WAREHOUSE-SERVICE-004

Actual consumption/scrap/rework від production має завершуватися коректним physical writeoff у Warehouse.

Context: Катерогії setup waste, production scrap, test samples, destroyed WIP треба розрізняти.

RECOVERED · HI-WAREHOUSE-SERVICE-005

Guided stock checks і discrepancy capture можуть виконуватися через Operations Assistant.

Context: Physical observation має evidence.

AGREED · HI-WAREHOUSE-SERVICE-006

Одиниці виміру повинні бути нормалізовані, щоб порівнювати expected vs actual.

Context: Інакше variance не має сенсу.

AGREED · HI-WAREHOUSE-SERVICE-007

A production reprint caused by internal defect must generate traceable additional physical material consumption/writeoff without silently changing the original planned quantity.

Context: OCR provides reprint execution context; Warehouse remains physical stock truth.

Roadmap: WH-REPRINT-01

AGREED · HI-WAREHOUSE-SERVICE-008

Legacy filename material tokens are parser candidates; Warehouse consumes canonical material IDs after Library/Calculator normalization.

Context: Do not make warehouse stock truth depend on a misspelled filename token.

Roadmap: WH-FILE-01

AGREED · HI-WAREHOUSE-SERVICE-009

Warehouse must reserve stock for active orders and calculate projected available stock after reservations, not only physical on-hand quantity.

Context: Reservation can push projected stock below the replenishment threshold before physical consumption occurs.

Roadmap: WH-STOCK-01

AGREED · HI-WAREHOUSE-SERVICE-010

Each controlled material needs a stock policy including at least minimum/reorder threshold and a normal replenishment quantity or target so Warehouse can raise a structured replenishment need.

Context: Financial feasibility and purchase execution remain separate ownership questions to reconcile.

Roadmap: WH-REPLENISH-01

AGREED · HI-WAREHOUSE-SERVICE-011

Warehouse must support controlled write-off/adjustment reasons for damaged, spoiled or otherwise unusable materials while preserving traceability of the stock movement.

Context: Storage damage and production-related material loss must not disappear from stock without evidence.

Roadmap: WH-ADJUST-01

Module — production_runtime_inspector

Intent count: 12

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-001

Production Runtime Inspector збирає machine and runtime telemetry, але не визначає бізнес-статус замовлення.

Context: Він є technical evidence source.

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-002

Telemetry потрібна для actual cycle time, downtime, machine problems і виявлення аномалій.

Context: Operations Control Registry використовує її для planned-vs-actual view.

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-003

Broken machine або довгі простої не повинні автоматично 'навчити' систему, що погана швидкість є новою нормою.

Context: Norm change лише через proposal/review.

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-004

Runtime data треба корелювати з order_id/operation_id/machine_id/revision там, де це безпечно.

Context: Щоб факти не змішувалися між роботами.

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-005

При STOP_WORK critical state runtime integration не повинна запускати нові commands після gate check failure.

Context: Перед machine command перевіряється current order state.

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-006

Runtime Inspector records the concrete physical device instance actually used for an operation, not only a generic machine family.

Context: Needed for maintenance, quality and true machine-performance analytics.

Roadmap: PRI-DEVICE-01

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-007

Similar/replaceable printers may share a capability queue while runtime evidence still identifies the exact device used.

Context: Planning capability and actual telemetry are separate concerns.

Roadmap: PRI-DEVICE-02

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-008

Runtime Inspector must observe end-to-end execution traces: process start, module/tool calls, waits, durations, retries, outcomes and dependency blocking.

Context: Its focus is what the live system actually executed, not repository/project conformance.

Roadmap: RTI-TRACE-01

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-009

Runtime Inspector must expose repeated-call, retry, wait-time and bottleneck evidence so loops and asymmetric producer/consumer delays can be detected and analyzed.

Context: Example: one module repeatedly waiting on another should become visible as runtime evidence.

Roadmap: RTI-LOOP-01

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-010

Runtime governance must track separate configurable budgets for retries, repeated call signatures, inter-module hops, database/tool calls, AI usage/cost and timeout/TTL.

Context: Configured budget exhaustion must lead to stop/degraded/manual-review behavior rather than endless retries.

Roadmap: RTI-BUDGET-01

AGREED · HI-PRODUCTION-RUNTIME-INSPECTOR-011

Runtime observability should minimize sensitive payload exposure; identifiers, contract versions, status, timings, classifications/hashes and error evidence are preferred unless payload access is explicitly authorized.

Context: Observability must not create a new unrestricted data-leak surface.

Roadmap: RTI-DATA-01

PROPOSED · HI-FP-U92-20260903-010

Runtime evidence should include AI/tool calls, repeat patterns, tokens, latency, cost, retries, escalation and budget-runway signals; provisional context checkpoint/handoff thresholds remain policy-tunable rather than hard business truth.

Module — forprint_project_inspector

Intent count: 28

RECOVERED · HI-FP-PROJECT-INSPECTOR-001

Inspector — незалежний observer/auditor: знаходить repo/standards/knowledge/metadata/dependency drift.

Context: He є semantic owner.

RECOVERED · HI-FP-PROJECT-INSPECTOR-002

Цикл: Inspector detects -> local executor repairs/interprets -> Inspector rechecks -> Blueprint governs.

Context: Так зберігається separation of concerns.

AGREED · HI-FP-PROJECT-INSPECTOR-003

Inspector має незалежно перевіряти completion attestation проти pinned governance snapshot.

Context: Модуль не може сам собі остаточно засвідчити compliance.

AGREED · HI-FP-PROJECT-INSPECTOR-004

Findings треба ранжувати BLOCKING/HIGH/NORMAL/LOW і подавати в local maintenance queues.

Context: Не всі проблеми однаково критичні.

AGREED · HI-FP-PROJECT-INSPECTOR-005

Safe deterministic repairs можуть з'явитися пізніше, але semantic/business changes не повинні виконуватись Inspector-ом самостійно.

Context: Межа автоматизації має бути явною.

AGREED · HI-FP-PROJECT-INSPECTOR-006

Inspector має бачити drift між стандартом, з яким стартував крок, і current standard, але враховувати supported legacy revision policy.

Context: Не робити фальшиві FAIL лише через новішу ревізію.

AGREED · HI-FP-PROJECT-INSPECTOR-007

Project Inspector owns Semantic Consistency Audit as an internal capability; no separate semantic-audit ForPrint module is created.

Context: Keep the ecosystem small; Inspector remains independent and does not become domain owner.

Roadmap: PI-S1

AGREED · HI-FP-PROJECT-INSPECTOR-008

Incremental Inspector review must compare last_inspected_commit/checkpoint to the current target commit, not rely only on git status.

Context: Committed daytime changes must remain visible to the next audit.

Roadmap: PI-S1

AGREED · HI-FP-PROJECT-INSPECTOR-009

Anything that can be proven deterministically should be checked by scripts before involving an LLM.

Context: Schemas, revisions, indexes, tests, contract support and evidence are cheaper/more reliable mechanically.

Roadmap: PI-S2

AGREED · HI-FP-PROJECT-INSPECTOR-010

When interpretive policy/business/legal/customer-facing content changes, Inspector builds a bounded semantic audit package for an LLM worker.

Context: The worker receives changed content plus applicable authorities, contracts, Human Intent and explicit questions.

Roadmap: PI-S4

AGREED · HI-FP-PROJECT-INSPECTOR-011

The semantic LLM worker may ask Inspector for additional bounded context instead of guessing missing authority.

Context: Inspector can repack relevant documents and rerun the review.

Roadmap: PI-S5

AGREED · HI-FP-PROJECT-INSPECTOR-012

Semantic review triggers are risk-based, not diff-size-based; a one-line policy change can require immediate LLM review.

Context: High-risk categories include identity, payment, legal, customer-facing, production-safety and authority changes.
Roadmap: PI-S7

AGREED · HI-FP-PROJECT-INSPECTOR-013

A portfolio-wide semantic sweep should initially run about monthly and may move to quarterly/semiannual cadence as the architecture matures.

Context: Event-driven high-risk audits remain active even when full-sweep cadence decreases.
Roadmap: PI-S8

AGREED · HI-FP-PROJECT-INSPECTOR-014

Inspector semantic findings need a stable taxonomy covering contradiction, ambiguity, ownership violation, contract divergence, term drift, stale authority, Human Intent divergence, hidden defaults, orphan rules and cross-module behavior conflict.

Context: Findings must be structured rather than a vague AI opinion.
Roadmap: PI-S6

AGREED · HI-FP-PROJECT-INSPECTOR-015

Semantic findings must carry severity, confidence, evidence, affected modules and recommended resolution owner.

Context: The report must be usable by Blueprint/human governance.
Roadmap: PI-S6

AGREED · HI-FP-PROJECT-INSPECTOR-016

Inspector and its LLM worker detect, verify, compare, classify, report and route; they do not invent business rules or silently rewrite domain semantics.

Context: A suspected defect in an authority document is routed back to that authority owner.
Roadmap: PI-S10

AGREED · HI-FP-PROJECT-INSPECTOR-017

For legally meaningful content, LLM review performs issue spotting and contradiction analysis but does not declare material legal compliance confirmed.

Context: Final legal certainty remains a separate human/legal-review boundary.
Roadmap: PI-S10

AGREED · HI-FP-PROJECT-INSPECTOR-018

Inspector should consume a future machine-readable Cross-Module Invariant Registry maintained under Blueprint governance.

Context: The registry makes semantic checks more deterministic without giving Inspector semantic ownership.
Roadmap: PI-S3

AGREED · HI-FP-PROJECT-INSPECTOR-019

Inspector should detect shared-UI drift, stale component adoption, self-inventory inconsistency, duplicate capability implementations and cross-module semantic divergence.

Context: Inspector reports/routes; it does not own the underlying semantics.
Roadmap: PI-S11

AGREED · HI-FP-PROJECT-INSPECTOR-020

Project Inspector must audit repository cleanliness in Blueprint and other ForPrint repositories, including schema drift, generated-file drift, duplicate capability/semantic surfaces, orphan current documents, naming/folder deviations and uncontrolled temporary artifacts.

Context: Inspector detects and routes evidence; it does not become semantic owner of another module.
Roadmap: PI-CLEAN-01

AGREED · HI-FP-PROJECT-INSPECTOR-021

Project Inspector must detect candidate duplicate/equivalent functions and compare them with common fixtures, edge cases and invalid inputs to expose semantic divergence such as different price/rounding results.

Context: A candidate duplicate is not automatically the same semantic function; evidence is required.
Roadmap: PI-DUP-01

AGREED · HI-FP-PROJECT-INSPECTOR-022

When equivalent functionality exists in multiple modules, Inspector should report implementation quality, ownership and consolidation options but must not automatically merge or replace implementations.

Context: Target direction is one appropriate canonical owner/contract where semantics are truly shared.
Roadmap: PI-DUP-02

AGREED · HI-FP-PROJECT-INSPECTOR-023

Dormant scripts, functions, and documents are review candidates, not automatic deletion candidates; Inspector must use evidence such as usage, ownership, dependencies, documentation, fallback purpose and current roadmap relevance.

Context: Rare disaster-recovery or safety functionality can be important despite long periods without use.
Roadmap: PI-DORMANT-01

AGREED · HI-FP-PROJECT-INSPECTOR-024

Project Inspector must identify orphaned or misowned historical capabilities and route them for KEEP/MOVE/MERGE/REWRITE/DEPRECATE/REMOVE review with evidence.

Context: Old Calculator/Telegram functionality is a priority area for this audit.
Roadmap: PI-OWNERSHIP-01

AGREED · HI-FP-PROJECT-INSPECTOR-025

Project Inspector must support periodic and risk-triggered conformance audits and emit bounded findings with evidence, severity, responsible owner and recommended disposition.

Context: Cleanliness, duplicate standards, stale instructions and semantic conflicts are ongoing controls.
Roadmap: PI-AUDIT-01

AGREED · HI-INSPECTOR-U131-20260906-001

Inspector should resolve a question to shared semantics, filter modules through inventory/index evidence, then filter significant implementations through metadata and dependency relationships before deep source inspection.

Context: This reduces full-repository semantic guessing.
Roadmap: INSPECTOR-SEMANTIC-QUERY

AGREED · HI-INSPECTOR-U131-20260906-002

Inspector findings should distinguish exact canonical matches, known aliases, linked provisional implementations and AI-inferred matches so uncertainty is visible rather than hidden.

Context: Manual review should concentrate on inferred or conflicting cases.
Roadmap: INSPECTOR-SEMANTIC-CONFIDENCE

AGREED · HI-INSPECTOR-U131-20260906-003

Inspector should detect duplicate current implementations without lineage/coexistence justification and later support read-only impact and conformance queries over shared contracts.

Context: Inspector advises and verifies; it must not silently delete, merge or reassign semantic ownership.
Roadmap: INSPECTOR-INVENTORY-CONFORMANCE

Module — forprint_strategic_control_plane

Intent count: 6

PROPOSED · HI-FP-STRATEGIC-CONTROL-PLANE-001

Strategic Control Plane у зрілому стані може отримати частину portfolio-level planning/optimization, яку Blueprint тимчасово тримає зараз.

Context: Точний поділ ще не погоджений.

AGREED · HI-FP-STRATEGIC-CONTROL-PLANE-002

Стратегічні рішення — priority shifts, phase boundaries, destructive/security/cross-repo exceptions — залишаються human-controlled.

Context: Навіть при сильній автоматизації.

PROPOSED · HI-FP-STRATEGIC-CONTROL-PLANE-003

Може аналізувати cost/quality/lead-time/executor evidence і пропонувати балансування портфеля.

Context: Але не міняти roadmap authority без governance path.

PROPOSED · HI-FP-STRATEGIC-CONTROL-PLANE-004

Може підтримувати what-if planning і critical-path scenarios для кількох модулів наперед.

Context: Це кандидат на майбутній рівень, не поточний blocker.

GAP · HI-FP-STRATEGIC-CONTROL-PLANE-005

Не завершено рішення, які функції назавжди лишаться у Blueprint, а які перейдуть сюди/Inspector.

Context: Це відкрите архітектурне питання.

AGREED · HI-FP-U92-20260903-007

Strategic Control Plane is long-horizon decision intelligence for quantified scenarios, investments and strategic memory; strategic decisions remain human.

Module — forprint_integration_gateway

Intent count: 12

RECOVERED · HI-FP-INTEGRATION-GATEWAY-001

Gateway має бути тонким integration boundary: validation, normalization, routing, idempotency, correlation.

Context: Не бізнес-мозок.

RECOVERED · HI-FP-INTEGRATION-GATEWAY-002

Він не повинен зберігати client/order/material/payment truth.

Context: Canonical data лишаються в owning modules.

AGREED · HI-FP-INTEGRATION-GATEWAY-003

Manual intake після нормалізації в Calculator і Telegram/Website requests можуть проходити через Gateway тоді, коли з'явиться реальна runtime handoff потреба.

Context: Не реактивувати його лише заради архітектурної краси.

RECOVERED · HI-FP-INTEGRATION-GATEWAY-004

Errors мають бути typed/structurally stable і не приховувати bad payload silent fixes.

Context: Краще explicit validation error.

AGREED · HI-FP-INTEGRATION-GATEWAY-005

Semantic Retrieval є окремим сервісом і не повинен перетворювати Gateway на search/AI engine.

Context: Gateway транспортує запит, не вирішує semantic truth.

AGREED · HI-FP-INTEGRATION-GATEWAY-006

Gateway should validate and route runtime payloads against published ACTIVE contract revisions rather than owning the contract lifecycle itself.

Context: Contract Registry says what is agreed; Gateway applies that agreement in transport/runtime.

Roadmap: GW-CONTRACT-01

AGREED · HI-FP-INTEGRATION-GATEWAY-007

Gateway must not silently reinterpret Library/CRM/Accounting semantics when normalizing a payload.

Context: If a semantic mapping changes, it requires the appropriate authority/contract revision.

Roadmap: GW-CONTRACT-02

AGREED · HI-FP-INTEGRATION-GATEWAY-008

Integration Gateway must provide a canonical inter-module message envelope and validate contract/schema/version compatibility before routing a request or response.

Context: Internal module implementation can differ; the boundary format must remain controlled.

Roadmap: IG-CONTRACT-01

AGREED · HI-FP-INTEGRATION-GATEWAY-009

Invalid or incompatible payloads must be rejected/quarantined with visible evidence rather than silently guessed, reshaped or ignored.

Context: The operator needs an administrative view of invalid exchanges and why they failed.

Roadmap: IG-VALIDATE-01

AGREED · HI-FP-INTEGRATION-GATEWAY-010

Gateway integration control must include idempotency, bounded retries, dead-letter/quarantine behavior and audit/correlation evidence for failed or repeated inter-module exchanges.

Context: Transport reliability must not create duplicate business operations or hidden infinite loops.

Roadmap: IG-RELIABILITY-01

AGREED · HI-FP-INTEGRATION-GATEWAY-011

Gateway validates transport/contract boundaries but does not become universal business-semantic owner; domain modules validate their own semantics and Contract Registry controls contract lifecycle.

Context: Runtime Inspector observes execution; Project Inspector audits design/conformance.

Roadmap: IG-BOUNDARY-01

Shared resource concurrency must preserve persistent request state and explicit queue/wait/retry/degraded outcomes; TEMPORARILY_UNAVAILABLE is not equivalent to NOT_FOUND.

Module — forprint_identity_access_service

Intent count: 8

AGREED · HI-FP-IDENTITY-ACCESS-001

ForPrint should use one shared Identity & Access Service instead of separate authentication inside Website, Calculator, CRM and Mobile.

Context: The module existed as a prior candidate and is now explicitly confirmed for formation.
Roadmap: IAM-01

AGREED · HI-FP-IDENTITY-ACCESS-002

Identity, authentication and authorization are distinct concerns and must remain explicit in architecture.

Context: Business relationship truth remains outside credential/session ownership.
Roadmap: IAM-01

AGREED · HI-FP-IDENTITY-ACCESS-003

Access control must support role defaults plus explicit per-user permission additions or restrictions.

Context: A manager can receive extra Production/Logistics access without a duplicate person.
Roadmap: IAM-03

AGREED · HI-FP-IDENTITY-ACCESS-004

Phone, email and Telegram are identifiers/contact paths, not immutable primary database keys.

Context: Use stable internal Account/Person IDs.
Roadmap: IAM-02

AGREED · HI-FP-IDENTITY-ACCESS-005

Cross-client/cross-organization access is deny-by-default and requires explicit scoped permission.

Context: Representing multiple organizations does not grant universal access.
Roadmap: IAM-04

PROPOSED · HI-FP-IDENTITY-ACCESS-006

Mature Identity should support sessions/devices, recovery, revocation, MFA/passkeys and auditable security events.

Context: Synthetic full-horizon capability for owner review.
Roadmap: IAM-05

PROPOSED · HI-FP-IDENTITY-ACCESS-007

Mature Identity should provide shared SSO/session semantics across ForPrint web/internal/mobile surfaces.

Context: Exact protocols remain an implementation decision.
Roadmap: IAM-07

AGREED · HI-FP-IDENTITY-ACCESS-008

External provider passwords/API keys are not user identity and belong to centralized secrets infrastructure.

Context: Identity may authorize access to secret-backed actions but does not own the external secret.
Roadmap: IAM-BOUNDARY-01

Module — logistics_service

Intent count: 12

RECOVERED · HI-LOGISTICS-SERVICE-001

Logistics Service — provider-neutral owner delivery/tracking truth.

Context: Telegram/CRM показують його стани, але не дублюють provider logic.

RECOVERED · HI-LOGISTICS-SERVICE-002

Taxi/local delivery має окремий provider contract, quote/request/assigned/in-progress/completed/cancelled states.

Context: Money-spending booking потребує human approval boundary.

RECOVERED · HI-LOGISTICS-SERVICE-003

Provider substitution/fallback не повинен змінювати channel contract.

Context: Інші модулі не мають знати специфіку кожного перевізника.

RECOVERED · HI-LOGISTICS-SERVICE-004

Retry/backoff/circuit breaker і reconciliation provider state vs local canonical state потрібні для надійності.

Context: Події delay/exception можуть бути proactive.

AGREED · HI-LOGISTICS-SERVICE-005

Order cancellation/stop-work і delivery cancellation — пов'язані, але окремі workflows; Operations Control Registry ініціює business stop, Logistics керує delivery consequence.

Context: Не змішувати production і перевізника.

AGREED · HI-LOGISTICS-SERVICE-006

Logistics Service roadmap deepening must be recovery-first because substantial prior design evidence already exists; only genuine gaps should be filled with PROPOSED synthesis before the next portfolio review.

Context: Do not replace the detailed existing Logistics design with a generic rewrite.

Roadmap: LOG-RECOVERY-01

AGREED · HI-LOGISTICS-U129B-20260905-001

Logistics is the first reference module for validating module-local self-knowledge, inventory/index maintenance, fresh-worker continuity and standardized completion reporting.

Context: Successful behavior becomes a reusable portfolio reference after explicit review.

Roadmap: LOG-H10-REFERENCE

AGREED · HI-LOGISTICS-U129B-20260905-002

The first substantial Logistics pool requires manual operator review after every prompt before another prompt is released.

Context: This is the calibration pool for later control-level decisions.

Roadmap: LOG-H10-REFERENCE

AGREED · HI-LOGISTICS-U129B-20260905-003

Each Logistics prompt should be executable by a fresh assistant from repository-owned AGENTS, inventory/index, prompt/contract/oracle, roadmap and prior completion evidence without requiring previous chat memory.

Context: Self-maintaining module knowledge is a prerequisite for reliable automation.

Roadmap: LOG-H10-BOOTSTRAP

AGREED · HI-LOGISTICS-U129B-20260905-004

After the first Logistics pool, an independent inventory/audit must verify the actual repository result before automation is widened or the reference profile is propagated.

Context: Compare claims, inventory, lineage, documentation, tests and repository reality.

Roadmap: LOG-H10-REFERENCE

AGREED · HI-LOGISTICS-U131-20260906-001

The Logistics H10 bootstrap should become the first practical reference for stronger module memory and lineage, but it should not be expanded into full retrospective metadata archaeology of every trivial helper before the first pilot pool can start.

Context: Establish significant implementation self-knowledge first, then grow coverage continuously.
Roadmap: LOG-H10-BOOTSTRAP

AGREED · HI-LOGISTICS-ARCH-20260919-001

Logistics retains authoritative shipment-specific workflow/provider/tracking truth while a parent or linked cross-domain Process Manager may consume typed shipment events, waiting conditions and outcomes.

Context: Recovered from the 2026-09-19 long-running-process discussion and reconciled with the current Logistics target-state direction.

Module — forprint_system_administration

Intent count: 11

AGREED · HI-FP-SYSTEM-ADMINISTRATION-001

System Administration відповідає за printer routing/fallback для job tickets та інших physical outputs.

Context: Business order truth лишається в Operations Control Registry.

AGREED · HI-FP-SYSTEM-ADMINISTRATION-002

Printer/machine availability та fallback мають бути технічними capability, а permission to choose/override — через Identity & Access policy.

Context: Не плутати routing з authorization.

AGREED · HI-FP-SYSTEM-ADMINISTRATION-003

Critical alerts потребують надійної audio/visual delivery infrastructure і escalation timers.

Context: Але semantic state STOP належить OCR.

PROPOSED · HI-FP-SYSTEM-ADMINISTRATION-004

Може підтримувати service health, local device inventory, deployment/runbook surfaces для shop-floor systems.

Context: Це інфраструктурний шар, не business orchestration.

AGREED · HI-FP-SYSTEM-ADMINISTRATION-005

Temporary/report workspace policy має бути впроваджена технічно: окремі operator_exchange/reports, artifacts, work, diagnostics, backups.

Context: Щоб tmp корінь не перетворювався на смітник.

AGREED · HI-FP-SYSTEM-ADMINISTRATION-006

System Administration must distinguish physical device instances by stable asset/device identity, model and serial even when workers casually call several machines 'Epson'.

Context: Operational analytics must know which actual device performed work.

Roadmap: SYS-DEVICE-01

AGREED · HI-FP-SYSTEM-ADMINISTRATION-007

Production planning should distinguish required capability, eligible device group/queue and concrete device instance instead of encoding a printer serial in every human filename.

Context: This permits failover between similar machines.

Roadmap: SYS-DEVICE-02

AGREED · HI-FP-SYSTEM-ADMINISTRATION-008

System Administration owns physical routing/fallback for Job Ticket printing; Operations Assistant may request a reprint but does not own printer configuration.

Context: Keep human interface separate from infrastructure routing.

Roadmap: SYS-TICKET-01

AGREED · HI-FP-SYSTEM-ADMINISTRATION-009

System Administration owns the physical PostgreSQL platform operations and centralized secrets infrastructure.

Context: Domain modules retain semantic ownership; provider credentials do not live in Telegram/Calculator.

Roadmap: SYS-DATA-01

AGREED · HI-FP-SYSTEM-ADMINISTRATION-010

SysAdmin autonomous troubleshooting must follow an explicit allowlist of low-risk supported actions; unlisted or higher-risk physical/system interventions must escalate rather than being improvised.

Context: Internet-assisted guidance is acceptable only within approved low-risk troubleshooting classes.

Roadmap: SYSADMIN-SAFETY-01

AGREED · HI-FP-U92-20260903-004

System Administration should mature as a recovery-first, offline-capable technical operations helper with bounded voice/mobile/Telegram entry through controlled interfaces.

Module — forprint_contract_registry

Intent count: 18

PROPOSED · HI-FP-CONTRACT-REGISTRY-001

Contract Registry має зберігати versioned inter-module/public contracts і їх adoption/compatibility metadata.

Context: He implementation code.

AGREED · HI-FP-CONTRACT-REGISTRY-002

Roadmap steps і completion attestations повинні посилатися на конкретні contract revisions, коли вони є dependency.

Context: Щоб не було 'контракт десь оновили'.

PROPOSED · HI-FP-CONTRACT-REGISTRY-003

Потрібна видимість consumer/provider compatibility і deprecation windows.

Context: Особливо для Calculator/Library/Gateway/OCR flows.

AGREED · HI-FP-CONTRACT-REGISTRY-004

Human-intent ledger не замінює contract registry: людська теза пояснює навіщо, контракт описує точну machine boundary.

Context: Обидва рівні потрібні.

GAP · HI-FP-CONTRACT-REGISTRY-005

Ми ще не проговорили детально кінцевий набір contract-lifecycle функцій цього модуля.

Context: Портфоліо має показувати цю невідомість, а не заповнювати її синтетикою.

AGREED · HI-FP-CONTRACT-REGISTRY-006

The previous open Contract Registry lifecycle question is conceptually resolved: the module is a versioned registry of significant inter-module interface agreements.

Context: This supersedes the uncertainty expressed by HI-FP-CONTRACT-REGISTRY-005 without deleting that historical GAP.

Roadmap: CR-01

AGREED · HI-FP-CONTRACT-REGISTRY-007

Library owns reusable domain semantics; Contract Registry owns the versioned interface agreement that carries those semantics between modules.

Context: Contract Registry must not become Library.

Roadmap: CR-01

AGREED · HI-FP-CONTRACT-REGISTRY-008

Gateway is a runtime consumer/enforcer of ACTIVE contracts; it is not the canonical home of contract lifecycle.

Context: Contract Registry and Gateway remain separate responsibilities.

Roadmap: CR-01

AGREED · HI-FP-CONTRACT-REGISTRY-009

A contract record should track provider, consumers, revision, purpose, schemas/artifacts, semantic authority references, compatibility, validators, examples, adoption and migration lifecycle.

Context: Exact physical schema is implementation work, but this target information set is agreed.

Roadmap: CR-02

AGREED · HI-FP-CONTRACT-REGISTRY-010

Contract lifecycle should distinguish DRAFT, PROPOSED, READY_FOR_ADOPTION, ACTIVE, SUPPORTED_LEGACY, DEPRECATED, BLOCKED, REVOKED and RETIRED.

Context: Lifecycle details can be refined while preserving the distinction between candidate, active, legacy and unsafe revisions.

Roadmap: CR-04

AGREED · HI-FP-CONTRACT-REGISTRY-011

Contract Registry must classify breaking versus non-breaking revisions instead of assuming compatibility from a version string.

Context: A breaking revision cannot silently become active while required consumers are unprepared.
Roadmap: CR-05

AGREED · HI-FP-CONTRACT-REGISTRY-012

Each contract revision needs a provider/consumer adoption matrix and independent Inspector conformance evidence.

Context: Blueprint uses the matrix to prevent one module running ahead of dependencies.
Roadmap: CR-06

AGREED · HI-FP-CONTRACT-REGISTRY-013

Contracts should carry representative fixtures and edge-case test vectors so support can be proven, not merely declared in documentation.

Context: Inspector can execute common conformance suites.
Roadmap: CR-07

AGREED · HI-FP-CONTRACT-REGISTRY-014

Contract changes need impact analysis showing direct and important indirect affected modules.

Context: This feeds Blueprint dependency balancing and rollout planning.
Roadmap: CR-08

AGREED · HI-FP-CONTRACT-REGISTRY-015

Core interface meaning should remain transport-neutral where practical; HTTP/queue/internal-call adapters should not silently redefine business semantics.

Context: Transport belongs to runtime integration concerns.
Roadmap: CR-03

AGREED · HI-FP-CONTRACT-REGISTRY-016

Contract Registry records readiness/adoption but does not unilaterally activate breaking ecosystem semantics.

Context: At current maturity, Blueprint remains the activation governance point after required evidence.
Roadmap: CR-12

AGREED · HI-FP-CONTRACT-REGISTRY-017

The first full Contract Registry lifecycle pilot will be the Calculator to Operations Control Registry Job Specification contract.

Context: The Calculator target is not fully fixed, so exact pilot fields remain future detailed work.
Roadmap: CR-09

AGREED · HI-FP-CONTRACT-REGISTRY-018

The mature target includes safe deprecation/migration/retirement and a portfolio compatibility view, but late automation steps remain synthetic until proven by earlier pilots.

Context: Keep the full horizon visible without pretending late mechanics are already decided.
Roadmap: CR-14

Module — forprint_marketing_orchestrator

Intent count: 9

RECOVERED · HI-FP-MARKETING-ORCHESTRATOR-001

Marketing Orchestrator — майбутній AI-assisted модуль для campaigns/social/content providers.

Context: He blocker commercial core.

RECOVERED · HI-FP-MARKETING-ORCHESTRATOR-002

Може вести content calendar, briefs, multi-model generation, approval workflow, publishing і performance learning.

Context: Autopublish boundary ще треба погодити.

RECOVERED · HI-FP-MARKETING-ORCHESTRATOR-003

Provider/model routing має порівнювати cost/quality/speed для text/image/video.

Context: Не прив'язувати архітектуру до одного AI provider.

AGREED · HI-FP-MARKETING-ORCHESTRATOR-004

Leads і customer response context мають переходити в CRM через contract, а не жити лише в marketing tool.

Context: CRM — customer workflow owner.

GAP · HI-FP-MARKETING-ORCHESTRATOR-005

Перший production-complete channel/content slice ще не визначено.

Context: Потрібно окреме продуктове рішення.

AGREED · HI-FP-U92-20260903-008

Marketing Orchestrator is a distinct Brand & Content Operations capability for campaigns, content, brand consistency and governed publication, not CRM/customer truth.

RECOVERED · HI-MKT-CREATIVE-TOOLCHAIN-20260922-001

For repeated short-video production, ForPrint should normally standardize the operator workflow on one primary video generator at a time so prompt patterns, operator skill and repeatability can accumulate instead of fragmenting effort across several parallel tools; this operating preference must not hard-wire the architecture to one permanent provider.

Context: Recovered from the owner's explicit July SMM statement that he does not want to work with several generators at once and wants one tool to learn gradually. The preserved source does not contain a final owner acceptance of Google Veo or any other specific provider.

RECOVERED · HI-MKT-ACQUISITION-20260922-001

A newly launched ForPrint website should not rely on passive organic discovery alone: Marketing should deliberately seed an initial stream of real visitors so demand, usability and conversion behavior can be observed instead of waiting indefinitely for search traffic to appear.

Context: Recovered directly from the owner's June SMM cold-start problem: a new site may remain effectively invisible and therefore needs deliberate exposure to at least a narrow audience. Exact channel/budget are not fixed here.

RECOVERED · HI-MKT-ACQUISITION-20260922-002

The first ForPrint acquisition experiments should be targeted and selective rather than broad: start with audiences where a positive response is plausibly higher, including local proximity as a business hypothesis, and measure whether that locality assumption actually improves conversion before treating it as durable policy.

Context: Recovered from the owner's explicit agreement with a point/selective promo campaign and his reasoning that nearby print customers have a convenience advantage. The source contains no measured ROI proof.

Module — cloud_backup_manager

Intent count: 6

RECOVERED · HI-CLOUD-BACKUP-MANAGER-001

Cloud Backup Manager тримає backup inventory, status dashboard plan і operator runbook.

Context: Web admin не повинен тихо запускати destructive backup/restore.

RECOVERED · HI-CLOUD-BACKUP-MANAGER-002

Operator flow: preview -> execute -> verify -> report; guarded actions потребують terminal confirmation.

Context: Це вже зафіксовано в project roadmap.

RECOVERED · HI-CLOUD-BACKUP-MANAGER-003

Scheduler має бути disabled-by-default до окремої readiness/authorization фази.

Context: Не вмикати automation просто тому, що код готовий.

AGREED · HI-CLOUD-BACKUP-MANAGER-004

Blueprint portfolio має бачити backup readiness як dependency для критичних модулів, але не управляти backup execution напямую.

Context: Модуль лишається самодостатнім.

PROPOSED · HI-CLOUD-BACKUP-MANAGER-005

Human-intent ledger для Backup може зберігати операторські причини safety boundaries, щоб майбутній помічник не зняв їх як 'зайві обмеження'.

Context: Особливо корисно для restore/mirror режимів.

AGREED · HI-FP-U92-20260903-005

Cloud Backup Manager has three core responsibilities: off-site backup of prepared domain artifacts, controlled one-way corporate-resource replication, and retention/integrity/recovery evidence.

Module — verification_lab

Intent count: 3

AGREED · HI-FP-U92-20260903-011

ForPrint requires a separate Verification Lab that intentionally generates valid, invalid, edge, adversarial, concurrency, fault and recovery tests before production exposure.

AGREED · HI-FP-U92-20260903-012

Verification Lab is distinct from Project Inspector and Runtime Inspector: Inspector audits conformance, Runtime Inspector observes live facts, Verification Lab actively challenges behavior in an isolated Test Plane.

AGREED · HI-FP-VERIFICATION-LAB-20260925-CHANGE-AWARE-001

Verification Lab should enrich adversarial coverage primarily when functional evidence changes: inspect bounded change evidence, use AI to design or extend relevant scenarios, convert useful scenarios into versioned deterministic regression corpus, avoid repeatedly regenerating unchanged test strategy, run cheap targeted regression routinely, and perform broader resource-budgeted stress/soak/accumulated-data sweeps periodically in Test Plane/staging.

Context: This refines existing H07/H08 rather than creating a new testing model. Exact overnight windows remain infrastructure-dependent.
Roadmap: VERIFICATION_LAB_CHANGE_AWARE_ENRICHMENT_20260925